

Toolio

ME 1670 Introduction to Engineering Graphics and Visualization



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Naitik Mundra
Calvin Arnold

ME 1670: Introduction to Engineering Graphics

Instructor: Dr. Raghu Pucha Fall
2025

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Chapter 1

Project Ideation

Project Description

Inefficiency in workshops has become a growing problem in both personal shops and industrial workplaces. As tools become more affordable and accessibility to machinery increases, clutter increases. As a result, several types of tools and fasteners often go unsorted, creating a hazardous workspace. This is not only extremely dangerous for individuals seeking to grow their personal workshops, but it is also highly inefficient, as it wastes time individually sorting fasteners and tools. Ultimately, the goal is to design a toolbox that automates several of these processes, making the workplace more efficient and safer for individuals with large personal projects. The toolbox would have the ability to switch on, and when tools or fasteners are placed inside, it will automatically place them in the appropriate boxes, which are easily accessible.

Constraints and Requirements

- In order to be able to fit the systems that will automatically sort the tools, we have to assume that the toolbox is of a larger size as opposed to a small handheld toolbox.
 - The average cost of toolboxes used in larger personal workshops is from \$1300-\$2000.
 - Cost analysis of our design: ~\$2500
 - The average height of a larger toolbox is around 5ft.
 - We estimate a height of around 4ft for our design.
 - The toolbox should feature a way to organize items, which in most toolboxes involves drawers. □ We use box containers to sort tools and fasteners.
 - Toolboxes should be easily accessible for a simple user experience. □ Our open design provides easy access to all tools.

Description of Features & Part Distribution

Fastener Sorting Wheel Subassembly (Daniel Velazquez)

This subassembly will sort bolts, nuts, and washers, which are essential parts of a workspace. The key component is the wheel that rotates each fastener until it is deposited in the hole that fits its shape. They then go down a slide with the assistance of a conveyor that pushes them along the slide, in which they fall into their appropriate size (1/4', 1/2', 1/8', etc.). There are boxes to catch each type of fastener, which are conveniently located and easily accessible.

1. Washer Slide
2. Nut Slide

3. Bolt Slide
4. Rod
5. Pusher
6. Wheel
7. Outer Container
8. Outer Box
9. Inner Wheel
10. Inner Box
11. Funnel
12. Collar
13. Belt Support
14. Inner Belt Support
15. Belt Foot
16. Belt Cylinder
17. Belt Square
18. Belt Holder

Chassis Subassembly (Naitik Mundra)

This subassembly is key to connecting the various components together and providing functionality to provide vertical elevation changes for components to be sorted to different levels of sorting belts. This is achieved by the use of a hook and pulley system capable of over 10 feet of extension, giving it plenty of room for the 4.5 foot tall chassis. It also has fully functional castor wheels, allowing the toolbox to be repositioned easily. Finally, it contains the actual branches on which the conveyor belt is placed, and the baskets/tubs of parts that attach to it. On these branches is also a key function: a pusher mechanism that pushes things placed onto the conveyor into the aforementioned baskets.

1. Wheel axle
2. Wheel bolt
3. Wheel bush
4. Wheel post
5. Wheel tyre
6. Wheel cross (X) bracket
7. Stem

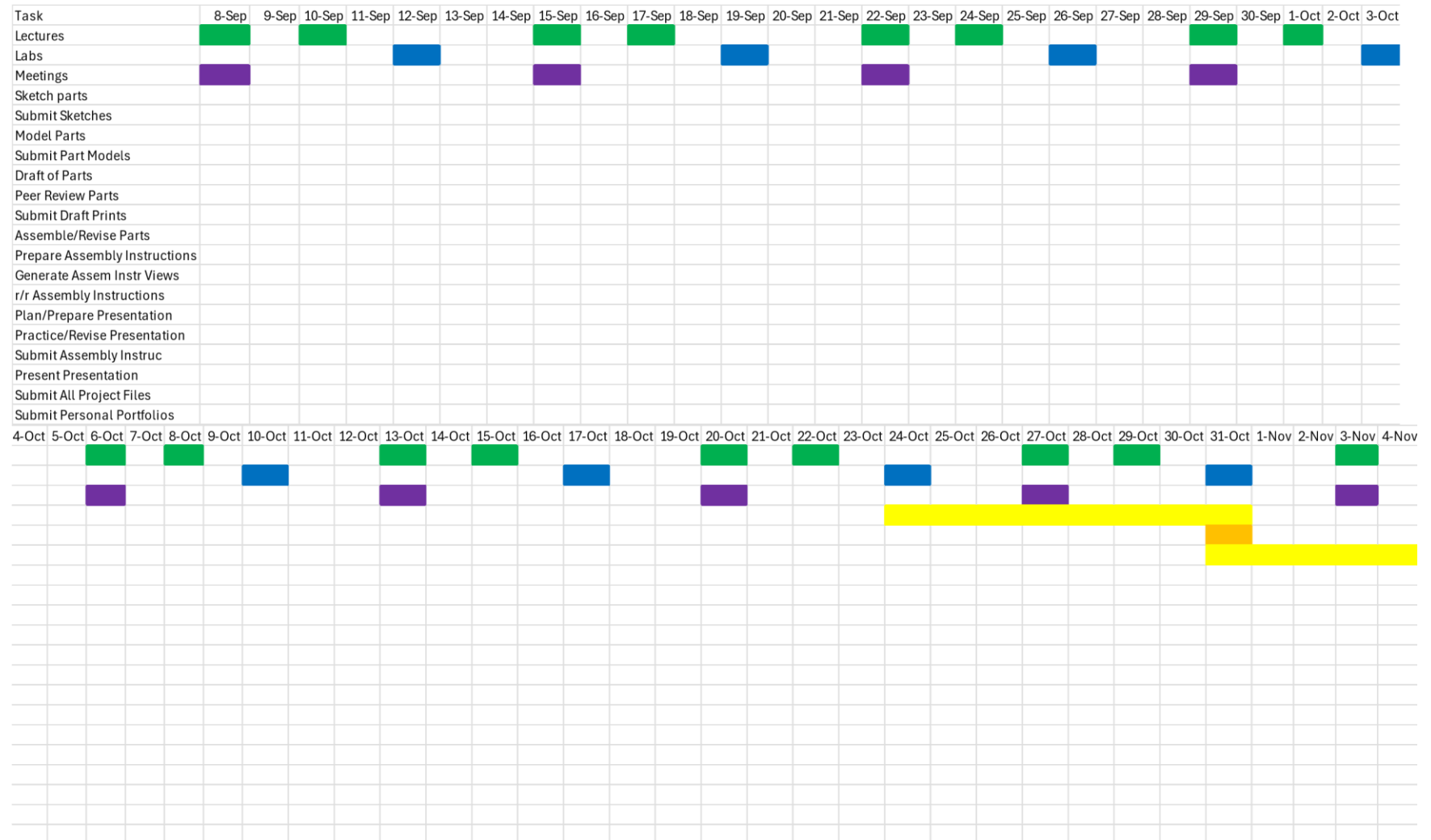
8. Branch
9. Pusher base
10. Pusher link 1
11. Pusher link 2
12. Pusher crank
13. Pusher lever
14. Pusher follower
15. Bucket
16. Pulley sheave
17. Pulley rope
18. Pulley hook

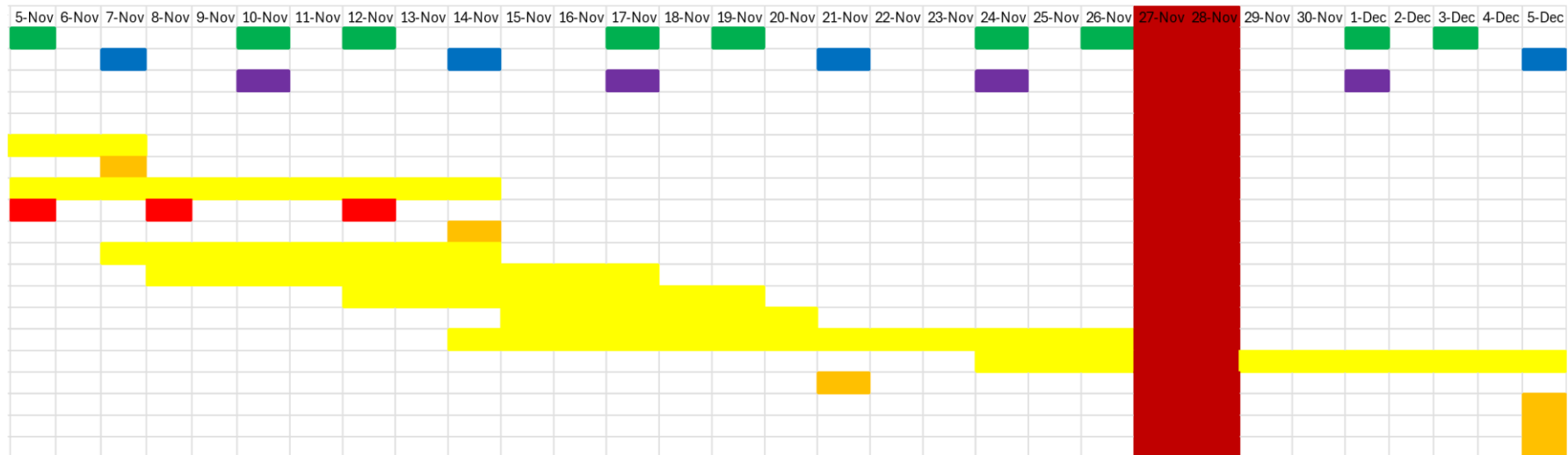
Conveyor (Calvin Arnold)

This subassembly distributes the parts from the central lift out to the bins to where they may be sorted. The horizontal transport of parts is important for the overall sorting mechanism. This is accomplished by a conveyor belt, of a chain type, that is constructed of a series of links out to the end of the belt. The belt is 12” long, allowing for transport of parts long distances.

1. Horizontal Guide Support
2. Conveyor Support
3. Conveyor Guide
4. Axle
5. Axle Pin
6. Conveyor Link
7. Conveyor Link Pin
8. Sprocket
9. Motor Mount
10. Motor

Project Management

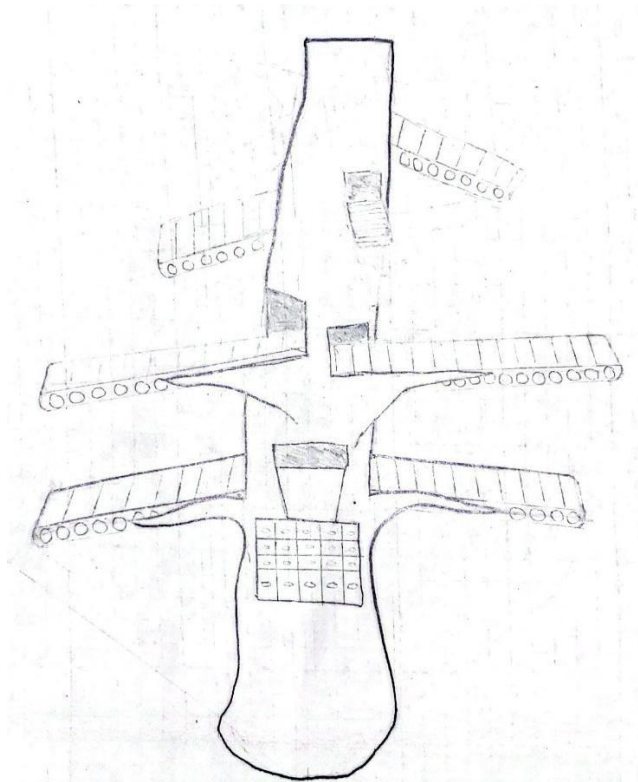




Chapter 2

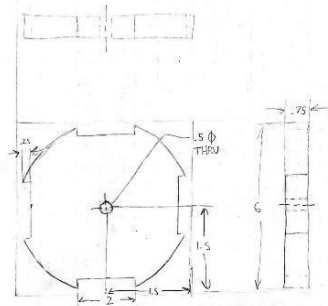
Preliminary Design

Initial 3D Contour Sketch

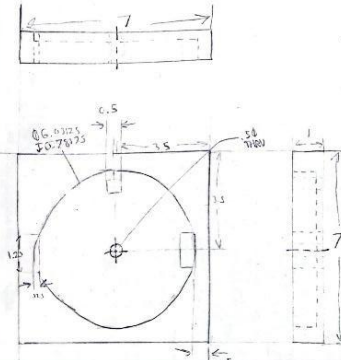


Hand Sketches: Daniel Velazquez

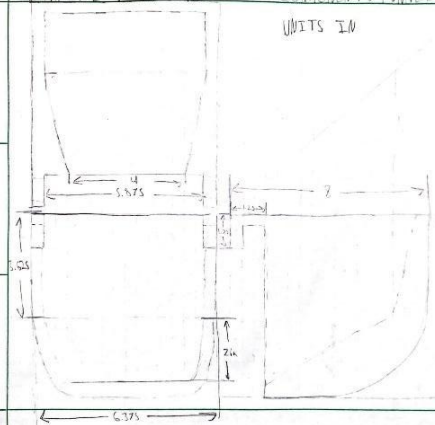
DANTE VELAZQUEZ COMPONENT 1: WHEEL
UNITS IN



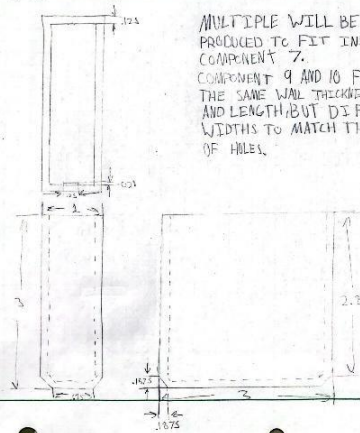
DANTE VELAZQUEZ COMPONENT 2: WHEEL COVER
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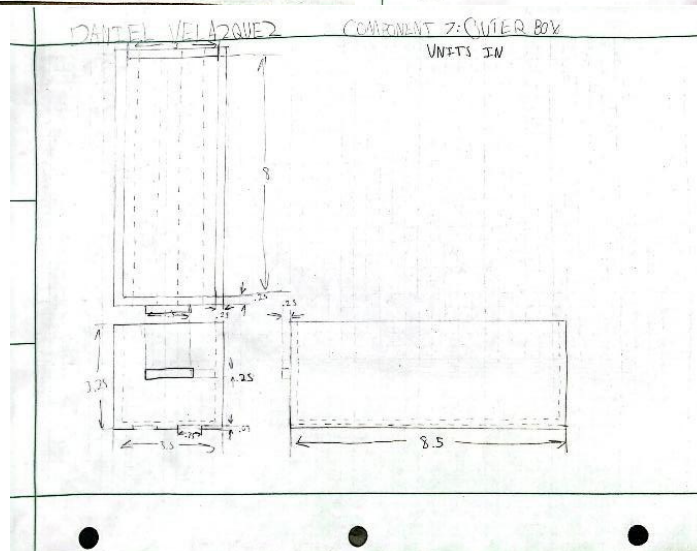
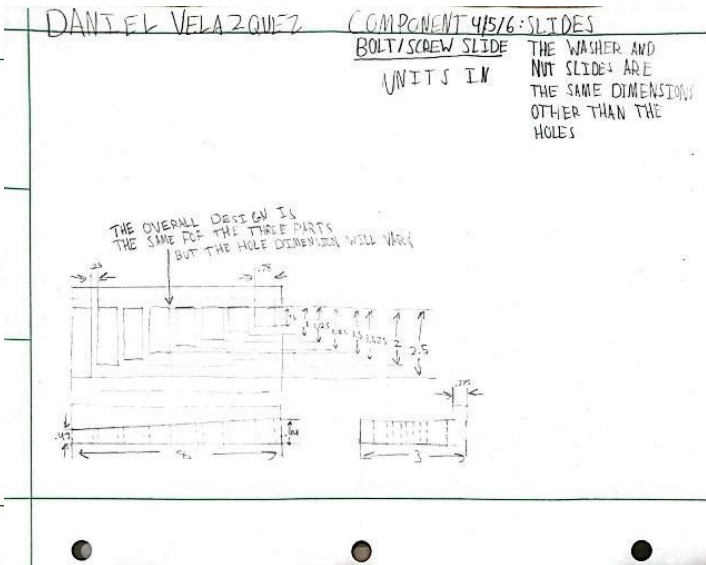
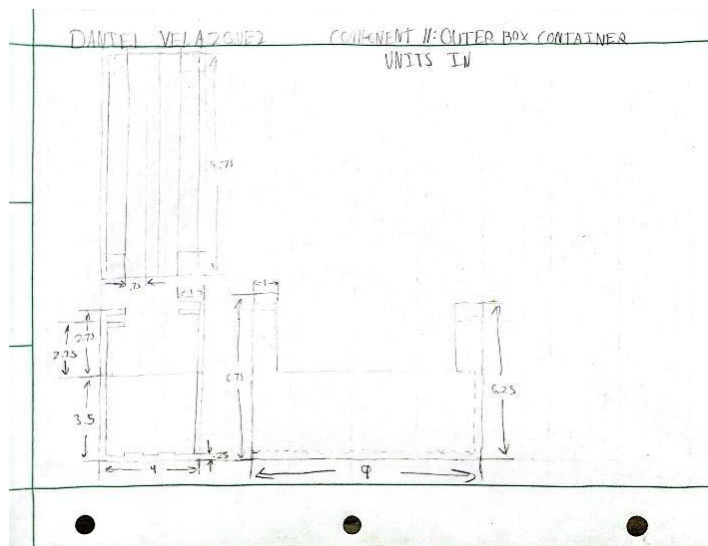
DANTE VELAZQUEZ COMPONENT 3: FUNNEL
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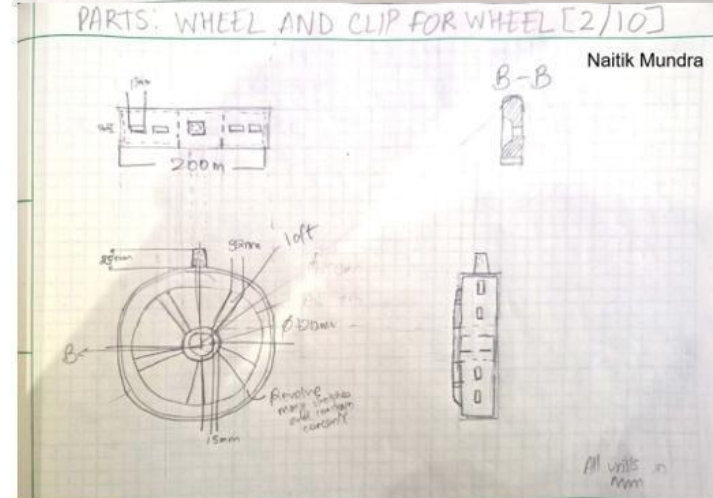
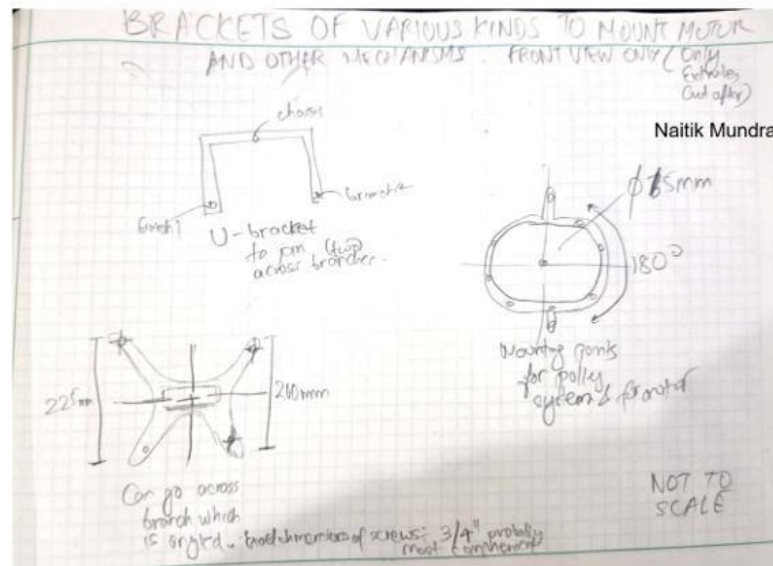
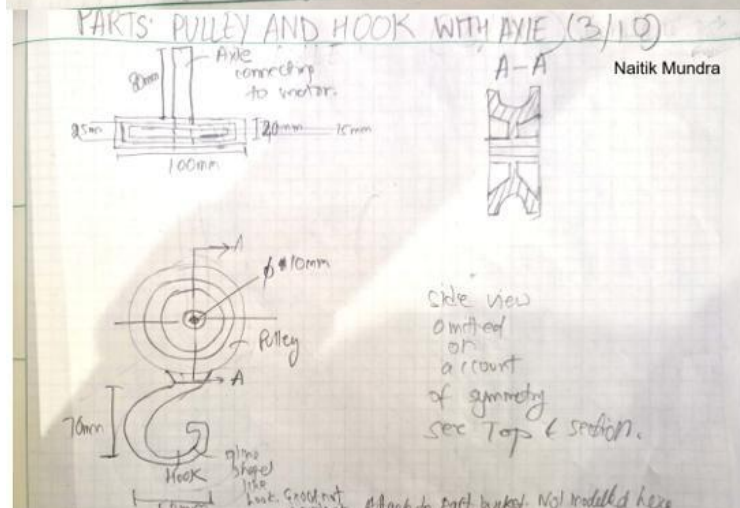
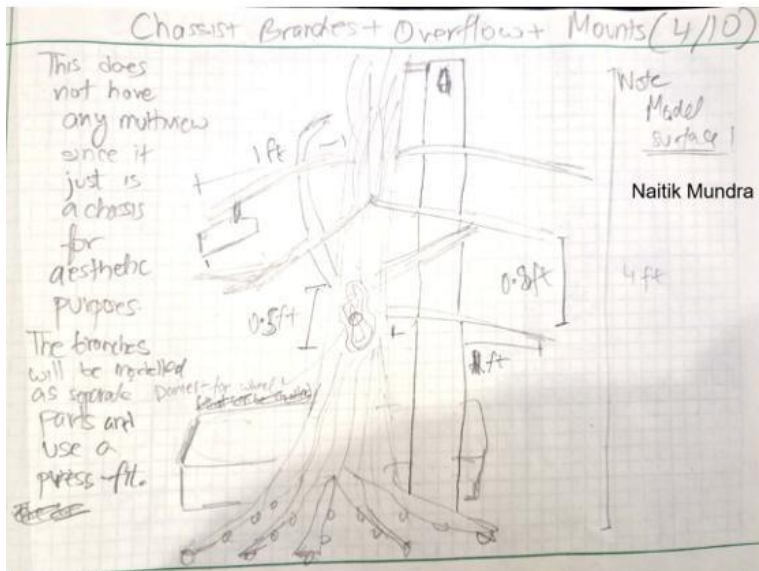
DANTE VELAZQUEZ COMPONENT 8: INDIVIDUAL BOXES
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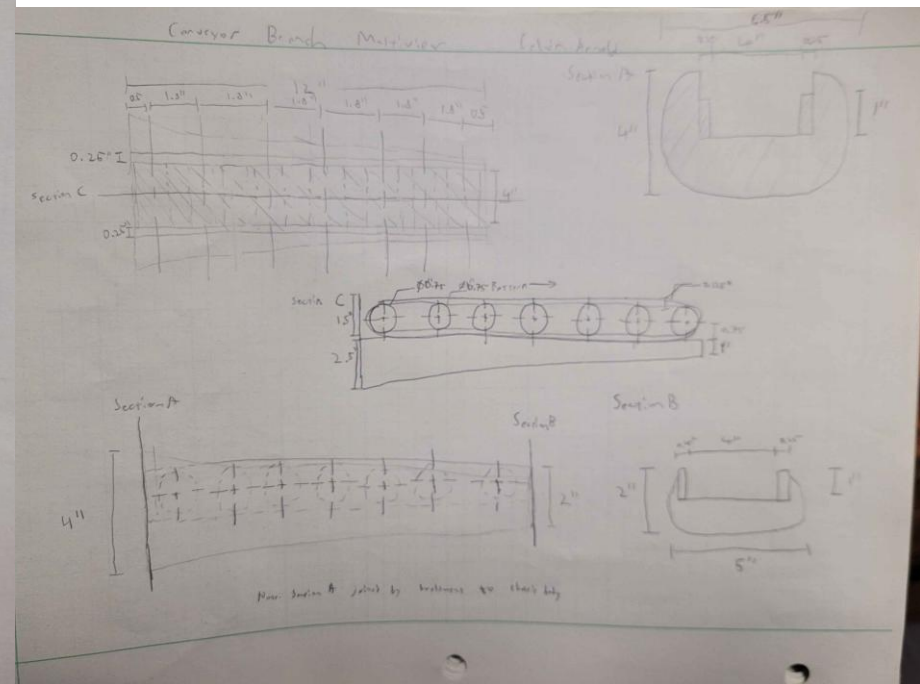
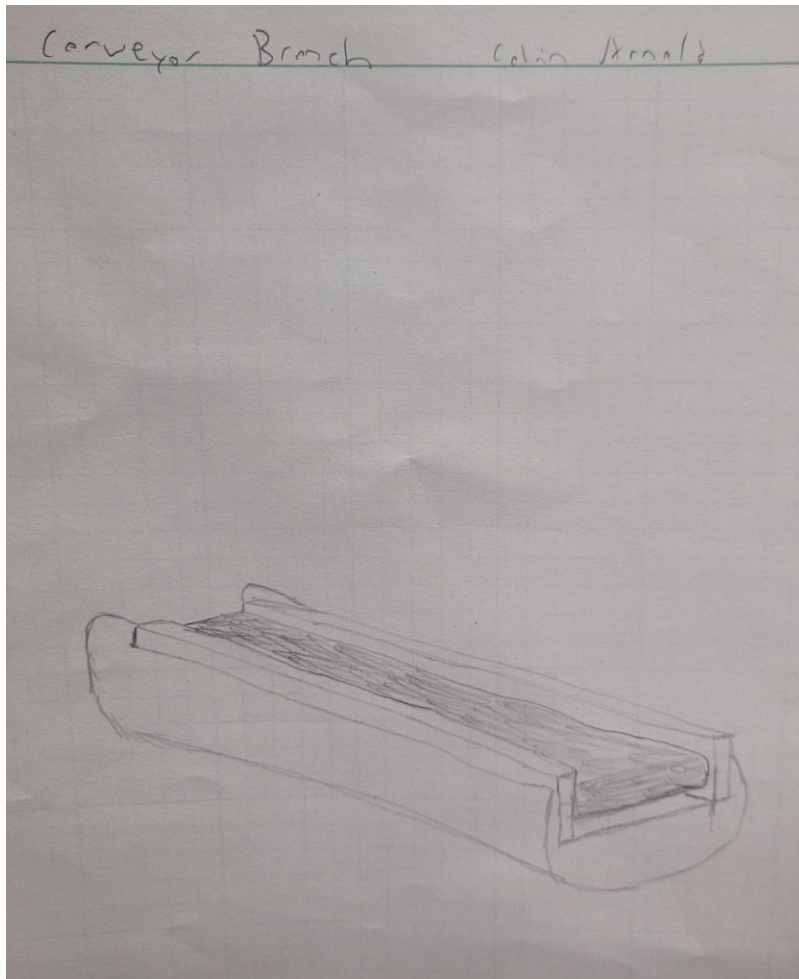
MULTIPLE WILL BE
PRODUCED TO FIT INSIDE
COMPONENT 7.
COMPONENT 9 AND 10 FEATURE
THE SAME WALL THICKNESS, HEIGHT,
AND LENGTH BUT DIFFERENT
WIDTHS TO MATCH THE NUMBER
OF HOLES.



Hand Sketches: Naitik Mundra



Hand Sketches: Calvin Arnold



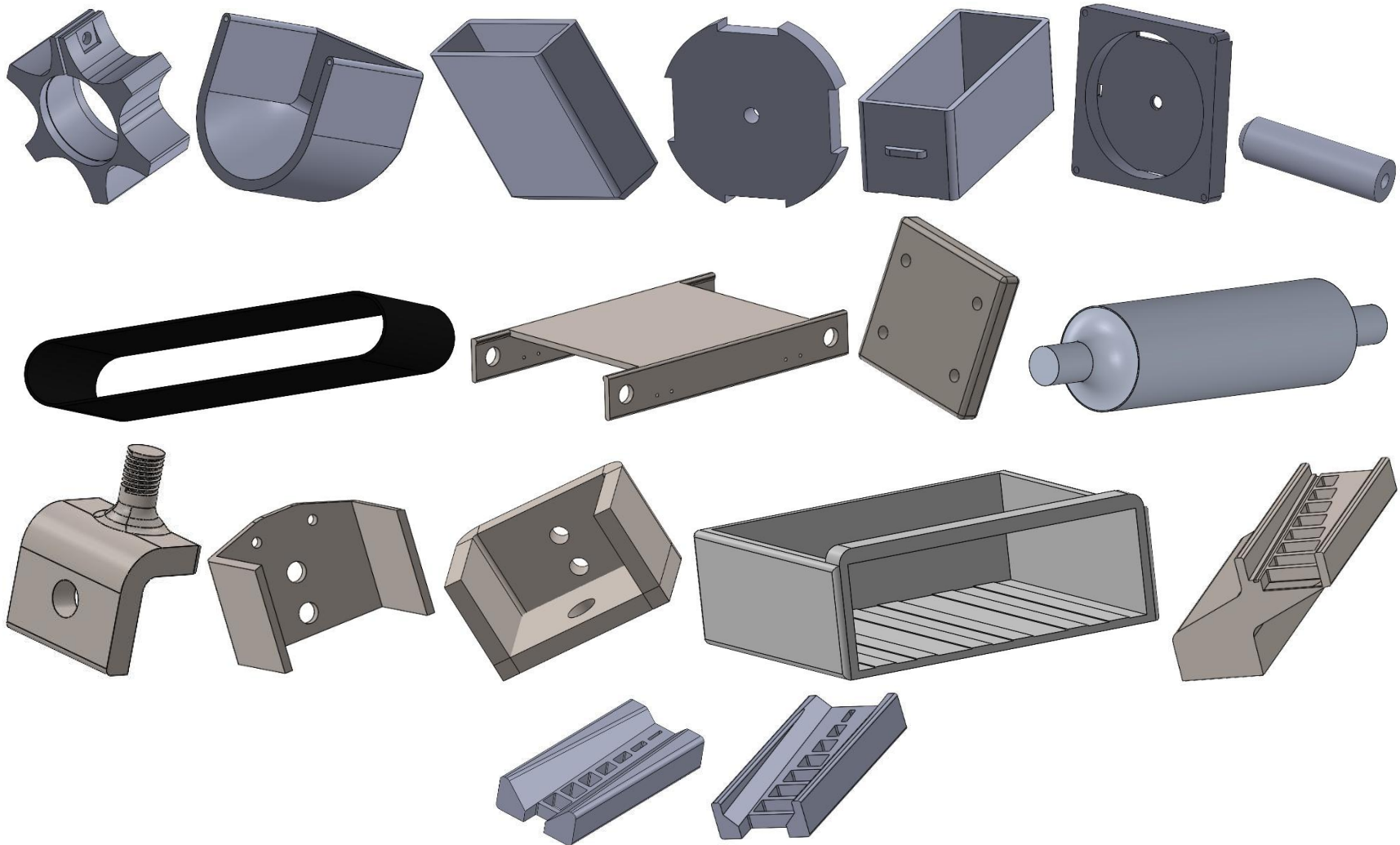
Chapter 3

Detail Design

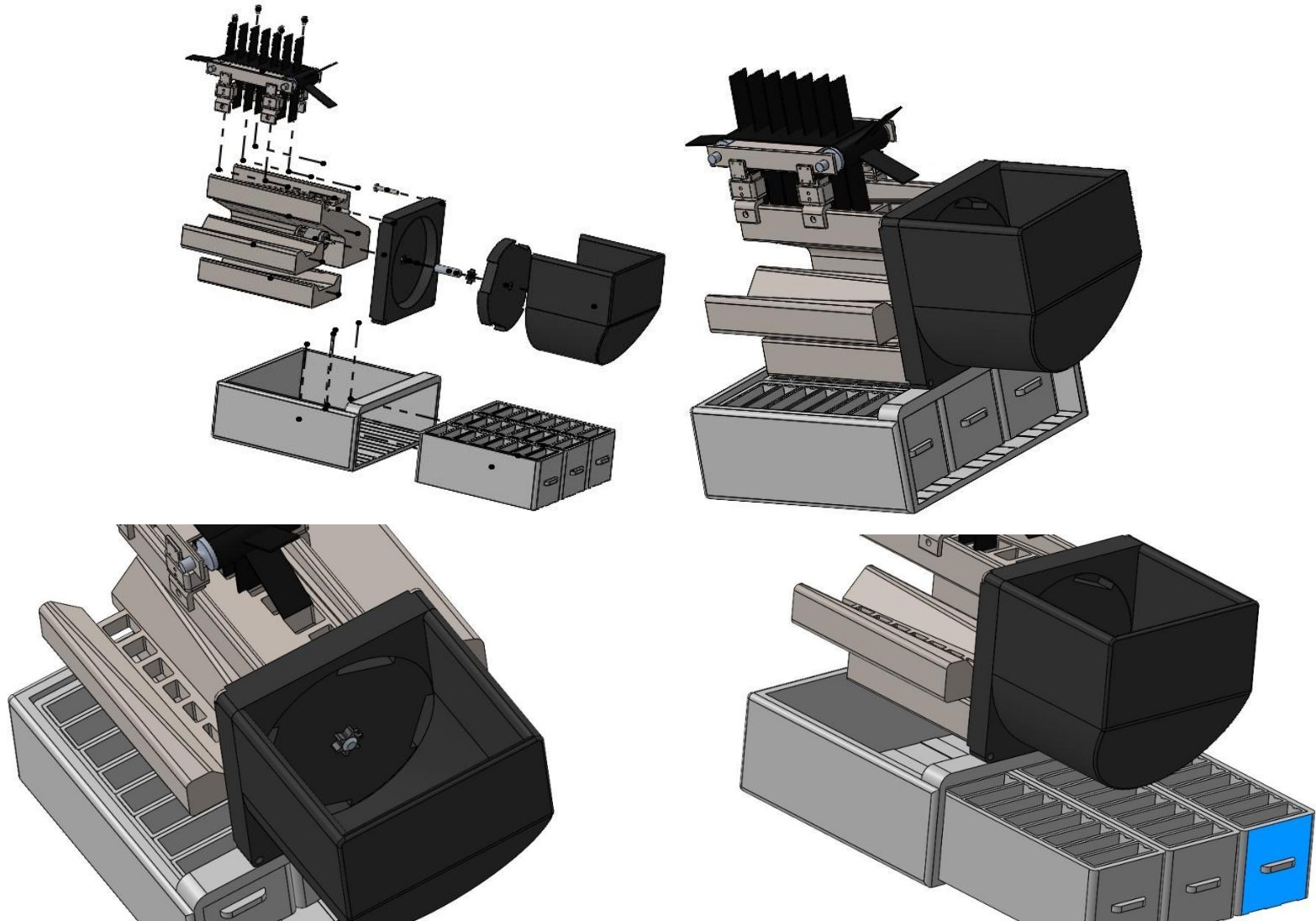
Final Assembly



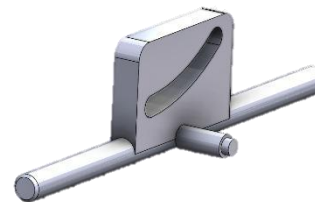
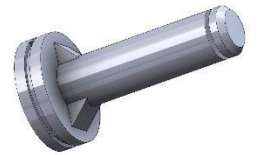
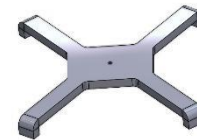
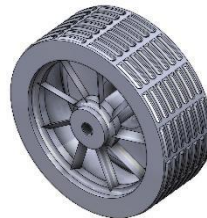
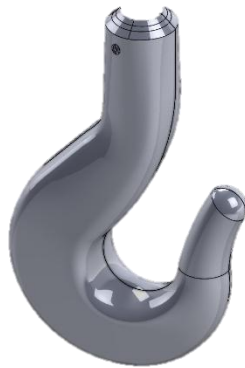
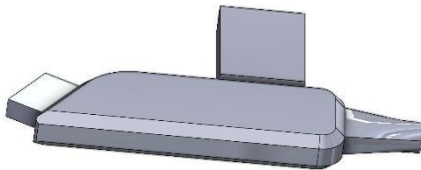
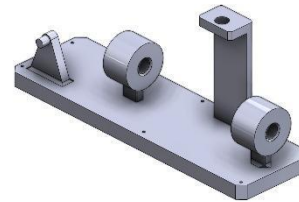
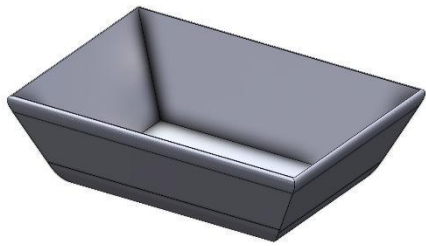
Part Models: Daniel Velazquez



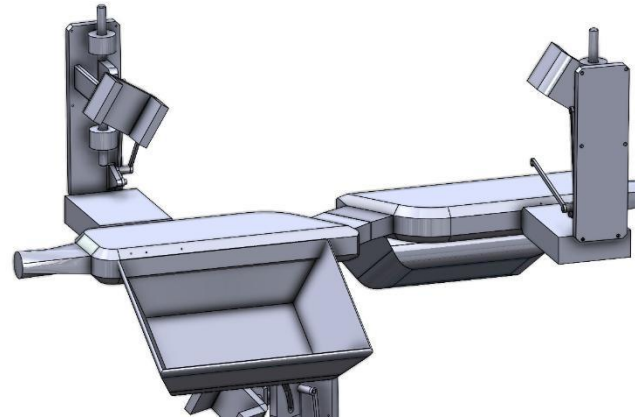
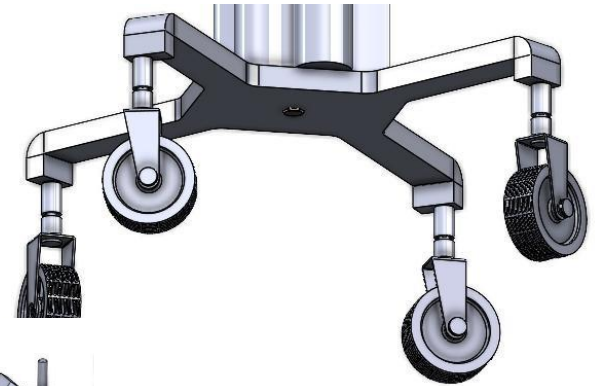
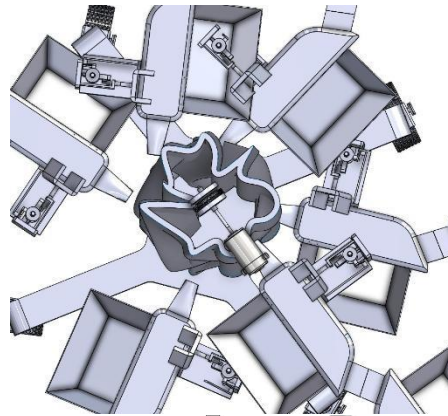
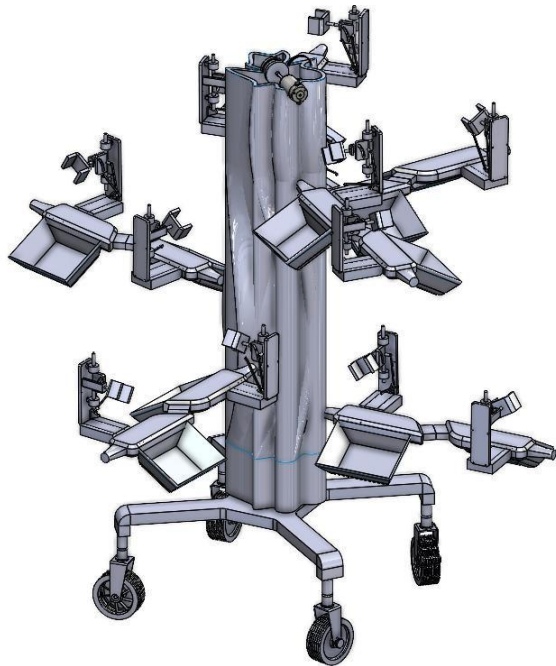
Subassembly: Daniel Velazquez



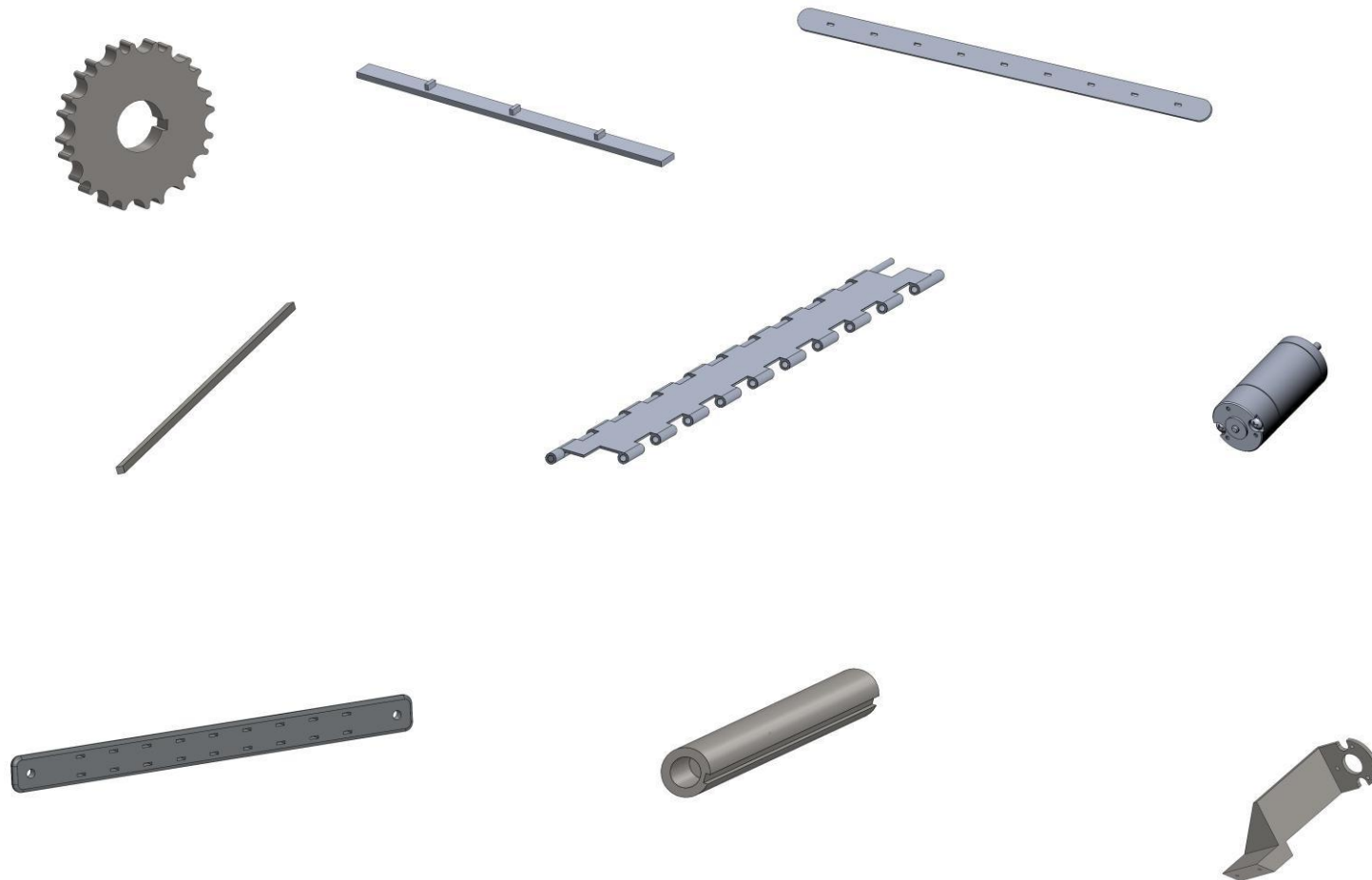
Part Models: Naitik Mundra



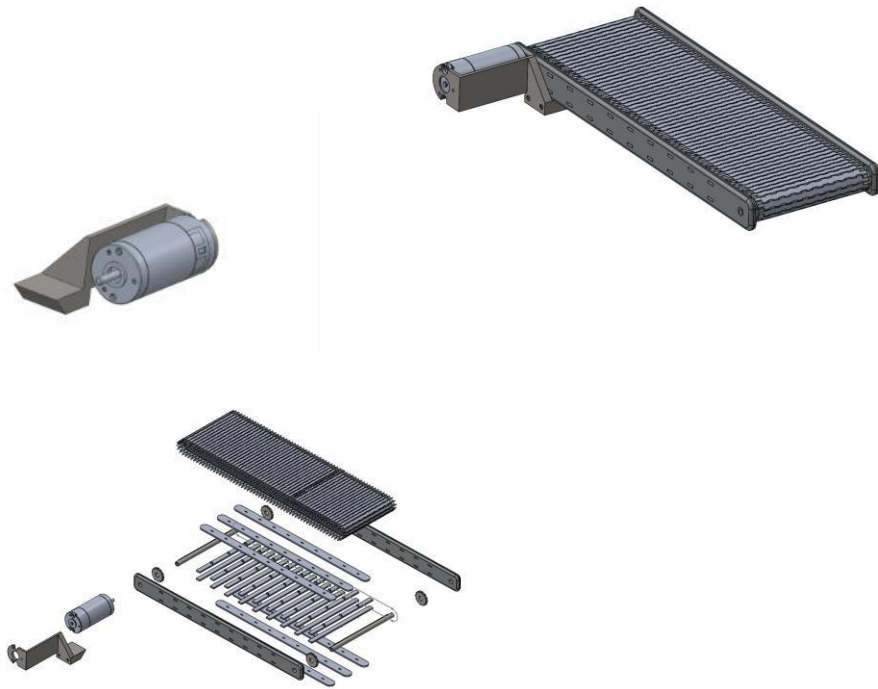
Chassis Subassembly: Naitik Mundra



Part Models: Calvin Arnold



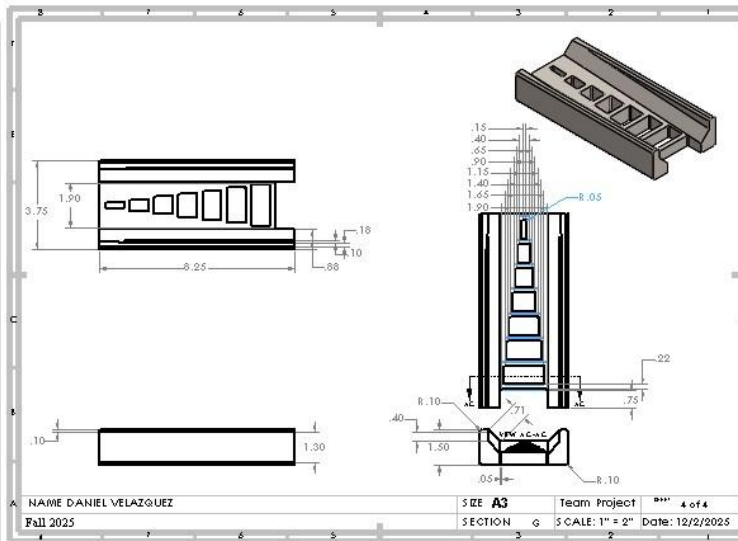
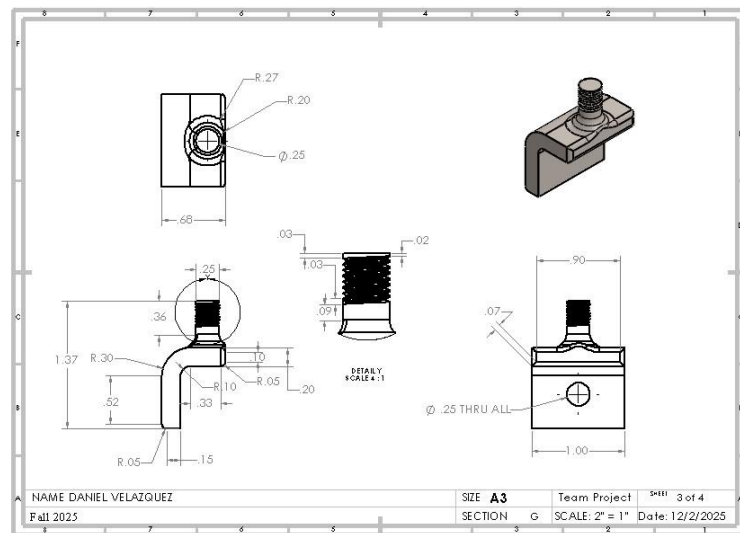
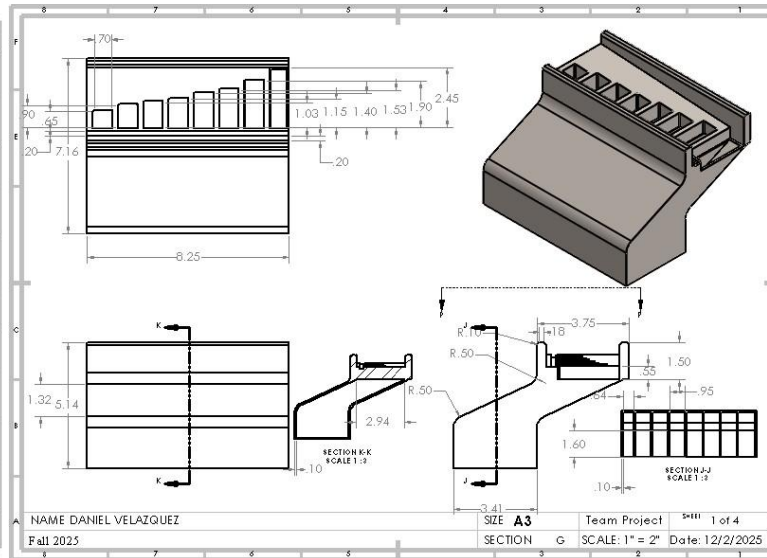
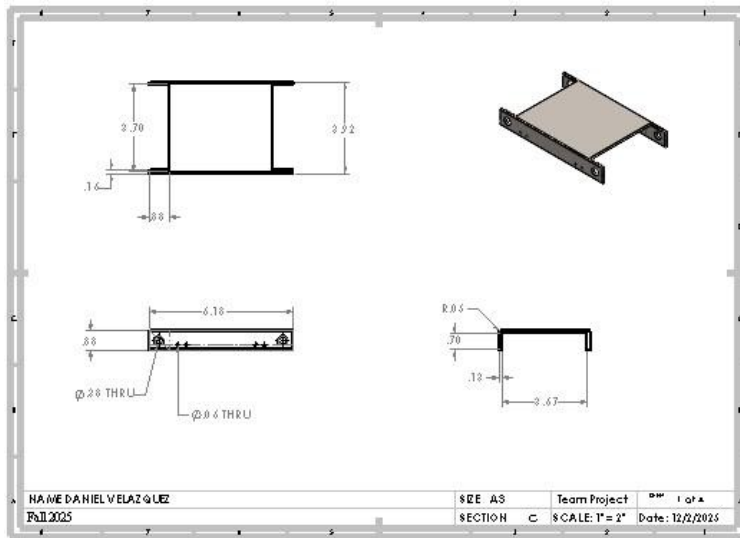
Subassembly: Calvin Arnold

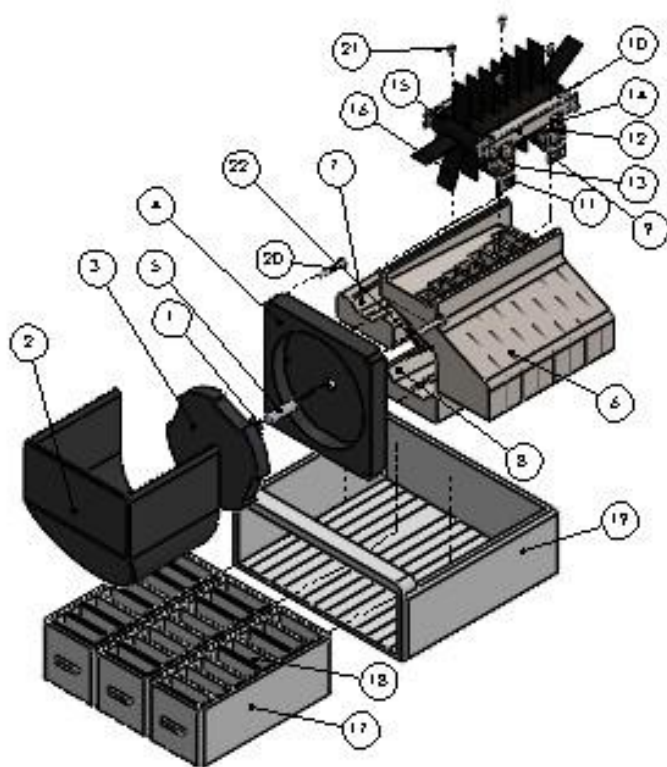


Chapter 4

Manufacturing Working Drawings

Working Drawings and Bill of Materials: Daniel Velazquez





ITEM NO.	PART NUMBER	DESCRIPTION	MATERIAL	QTY.
1	Collar	Collar used to hold rod in place	PP Copolymer	1
2	Funnel	Funnel to direct fasteners to wheel	PE High Density	1
3	Inner Wheel	Wheel that carries fasteners to slides	6061 Alloy	1
4	Outer Wheel	Outer wheel to contain inner wheel	6061 Alloy	1
5	Rod	Rod that rotates inner wheel	6061 Alloy	1
6	Slide-Bolt	Slide for bolts	201 Annealed Stainless Steel (SS)	1
7	Slide-Nut	Slide for nuts	201 Annealed Stainless Steel (SS)	1
8	Slide-Washer	Slide for washers	201 Annealed Stainless Steel (SS)	1
9	Belt Holder		ABS 1316 Stainless Steel Sheer (SS)	1
10	Belt Cylinder	Rotates the belt	6061 Alloy	2
11	Belt Pin	Connects the belt system to the slide	ABS 1316 Stainless Steel Sheer (SS)	1
12	Inner Support	Support for belt system	ABS 1316 Stainless Steel Sheer (SS)	1
13	Belt Support	Supports belt system	ABS 1316 Stainless Steel Sheer (SS)	1
14	Belt Square	Supports belt system	ABS 1316 Stainless Steel Sheer (SS)	1
15	Belt	Moves items along	Silicon Rubber	1
16	Pusher	Pushes items along slide	Silicon Rubber	17
17	Outer Box	Box that holds inner boxes	ABS	3
18	Inner Box	Boxes that contain fasteners	ABS	24
19	Outer Container	Allows access to fastener boxes	ABS	1
20	CR-BWMS D25-28x1.5x1-N			2
21	Sector head coll screw_01			1
22	6331 (1)	Compact Round-face DC Motor	201 Annealed Stainless Steel (SS)	1

NAME DANIEL VELAZQUEZ

SIZE **A3**

Team Project

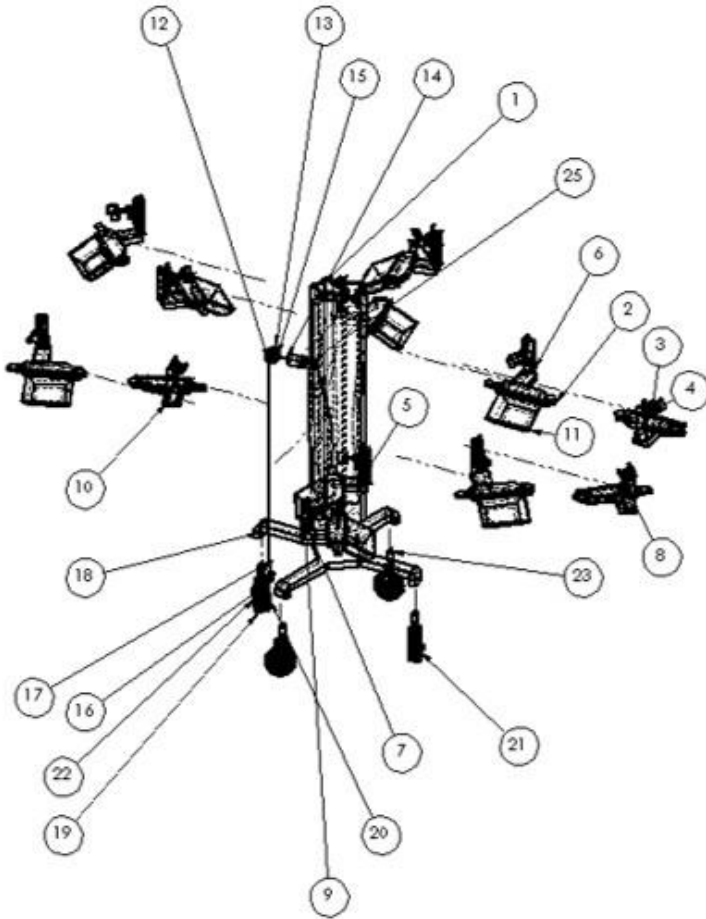
Sheet 1 of 1

Fall 2025

SECTION G

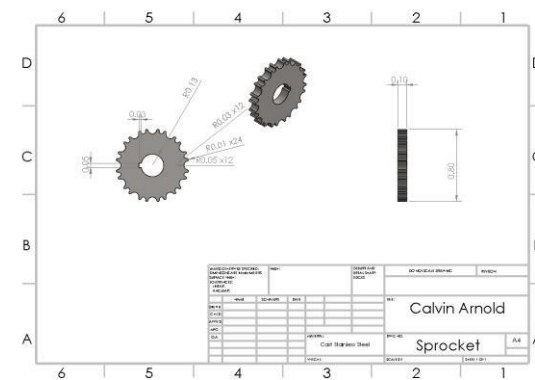
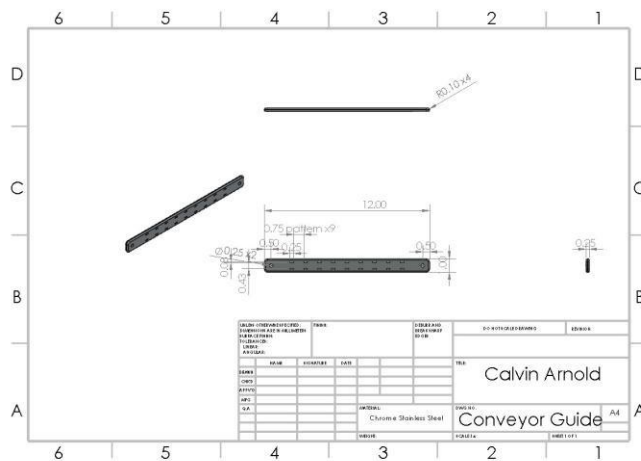
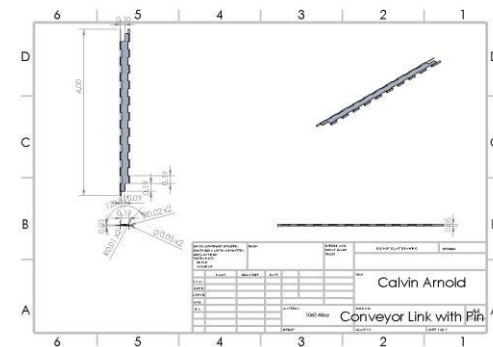
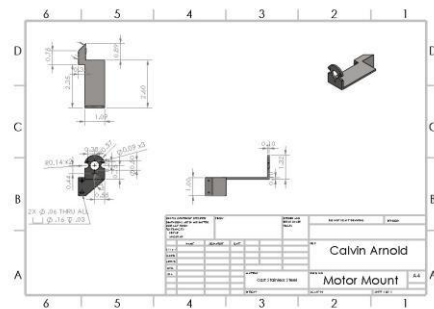
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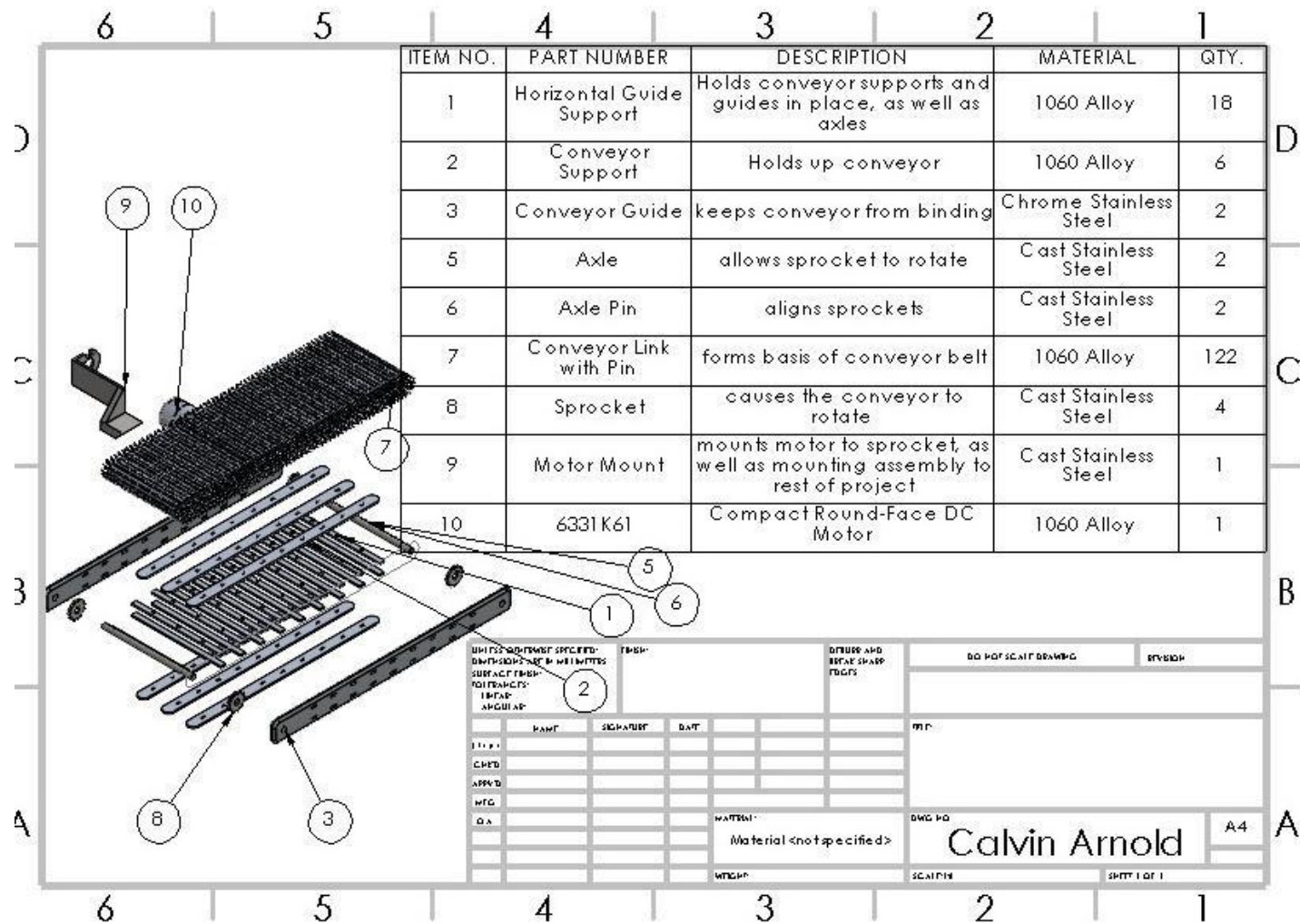
Date: 12/2/2025



ITEM NO.	PART NUMBER	DESCRIPTION	MATERIAL	QTY.
1	Mundra,Naitik_Stem	Main trunk	Teak	1
2	Mundra,Naitik_Bran ch	Where stuff goes. 2 on each side on each level	Teak	12
3	PusherBase	Base of pusher in branch	AISI 316 Stainless Steel Sheet (SS)	12
4	Follower	Cam mechanism	6061 Alloy	12
5	PusherLever	Crank force transfer	6061 Alloy	12
6	PusherLink	Crank force transfer	6061 Alloy	12
7	PusherLink2	Crank force transfer	6061 Alloy	12
8	Piston	Head to push object into bucket	AISI 316 Stainless Steel Sheet (SS)	12
9	PusherCrank	Crank to rotate to push	6061 Alloy	12
10	CR-BHMS 0.06- 80x0.5x0.5-N		AISI 316 Annealed Stainless Steel Bar (SS)	24
11	Bucket	Storage	PE High Density	12
12	Mundra,Naitik_Pulle y	Sheave	6061 Alloy	1
13	Rope^PulleyAndRo pe	Rope directly tensioned on sheave	Nylon 101	1
14	Mundra,Naitik_Axle	Connection between motor and pulley	AISI 316 Stainless Steel Sheet (SS)	1
15	Bushing^VerticalEle vator	For axle	201 Annealed Stainless Steel (SS)	2
16	Mundra,Naitik_Hook	Hook for objects to grapple to when moving levels	AISI 316 Stainless Steel Sheet (SS)	1
17	RopeExtension^Vert icalElevator	Rope that moves vertically	Nylon 101	1
18	Mundra,Naitik_Cross Bracket	4 slot bracket to connect wheels	AISI 316 Stainless Steel Sheet (SS)	1
19	Mundra,Naitik_Whe el	Tyre component	POLYURETHANE (11671)	4
20	Mundra,Naitik_Bolt	Lock for Wheel	AISI 316 Stainless Steel Sheet (SS)	4
21	Mundra,Naitik_Brac ket	Attach to posting	6061 Alloy	4
22	Mundra,Naitik_Bush	No axle	6061 Alloy	4
23	Mundra,Naitik_Post	Connect to cross bracket	AISI 316 Stainless Steel Sheet (SS)	4
24	HBOLT 0.7500- 16x1.125x1.125-N		201 Annealed Stainless Steel (SS)	1
25	6331 K13	Compact Round-Face DC Motor	Chrome Stainless Steel	1

Working Drawings and Bill of Materials: Calvin Arnold





Assembly Instruction Manual: Fastener Sorting Wheel Subassembly (Daniel Velazquez)

The subassembly can be broken into:

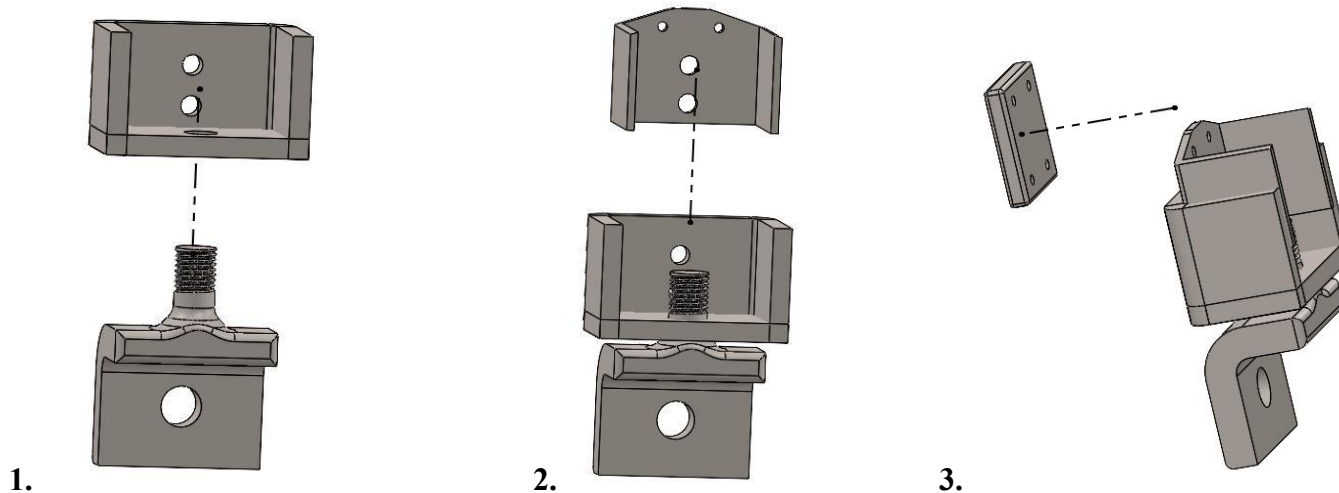
- Small Slide Conveyor

- Entire Subassembly

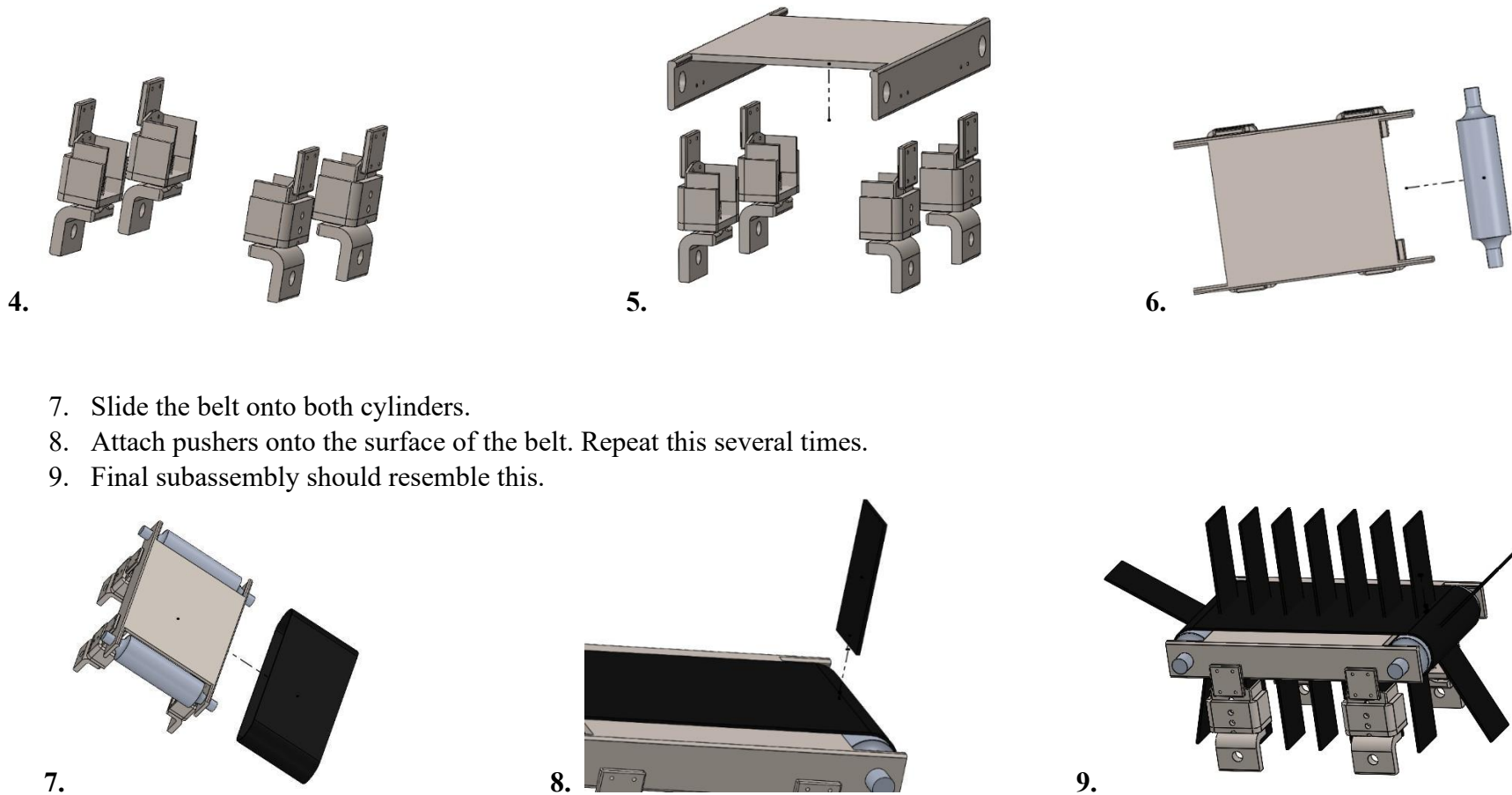
The entire subassembly is bolted to the main assembly.

Small Slide Conveyor

1. Align the bottom hole on the outer bracket with the thread of the belt foot. Orient them so that the face of the outer bracket that is hollow is parallel with the face of the belt foot. Push these together so that the thread is sticking out of the other side, and then tighten a nut (shown in the final subassembly) to hold them in place.
2. Align the bottom hole of the inner bracket with the top hole of the outer bracket and secure with a nut and bolt.
3. Use the bottom two holes of the square bracket and make them concentric with the two small holes on the top of the inner bracket. Ensure the square bracket is positioned on the side opposite the open face.

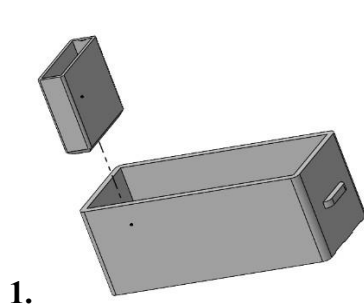


4. Repeat 4 times. Orient them so that two are next to each other, and two are directly opposite to those two.
5. The outer faces of the belt holder have small holes meant to align with the inner holes of the 4 small legs. Hold these in place using small nuts and bolts.
6. Slide the cylinder into place. The sides of the belt holes have two large holes to fit the small sides of the cylinder. Repeat this for the other side.

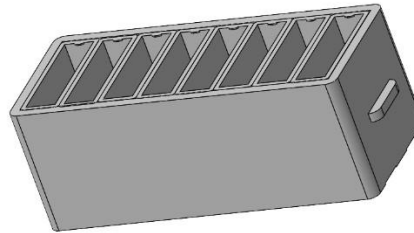


Entire Subassembly

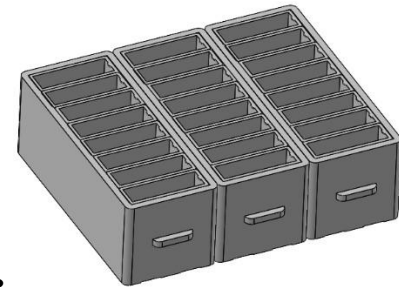
1. Place the inner box with the open side facing upwards inside the outer box. Orient is so that the long side of the inner box is touching the shorter face of the outer box.
2. Repeat this until there is no more space to fix small boxes.
3. Create three of these and align them with each side touching each other.



1.

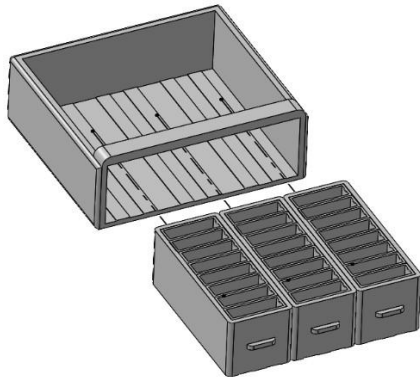


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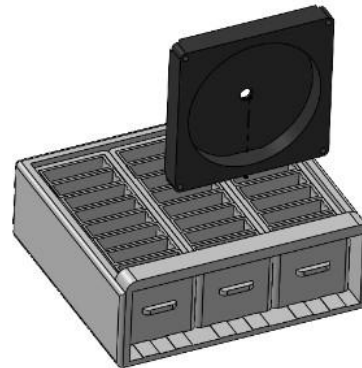


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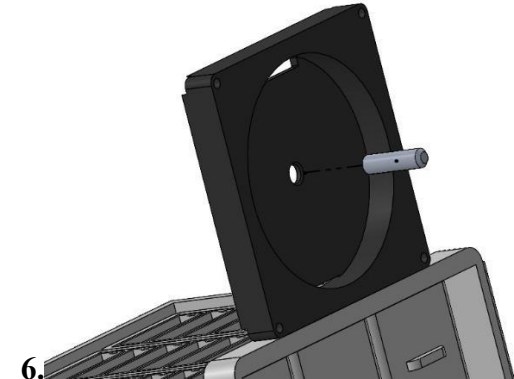
4. The outer container has three sets of 2 guiding lines for the bottom of the outer boxes to follow. After aligning them correctly, simply slide them into the until they touch the back wall with the handle facing towards the open side.
5. Use the outer wheel holder and place it at the midpoint of the top of the flat surface on the outer container. Weld these two together.
6. Slide the small rod with the rounded edge facing the direction of the open side of the outer wheel in the large hole in the outer wheel. Leave this for now.



4.

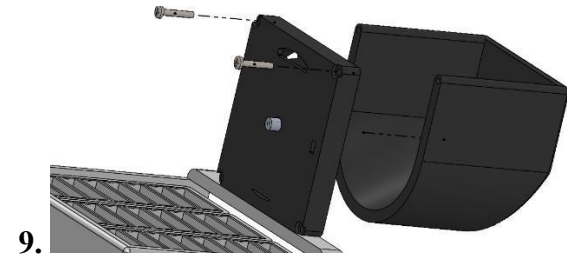
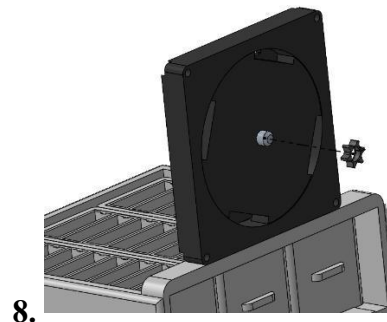
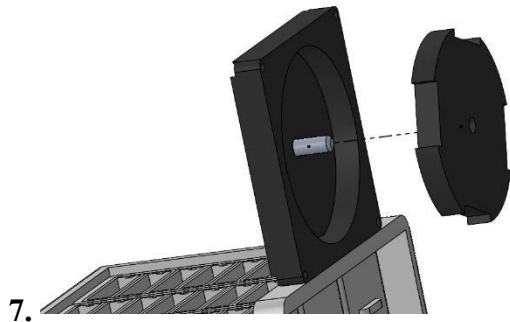


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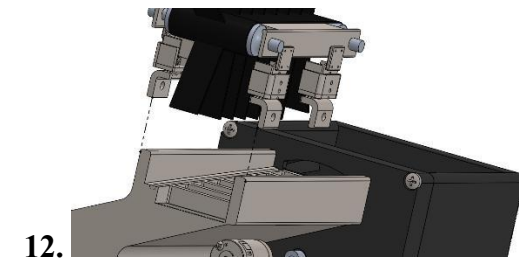
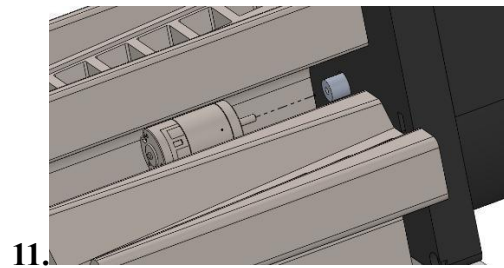
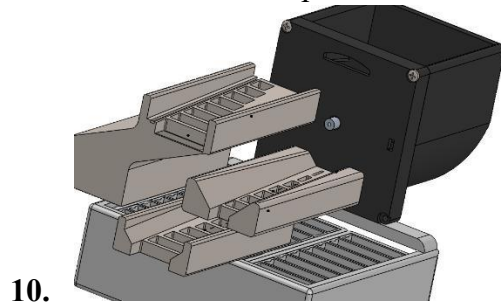


6.

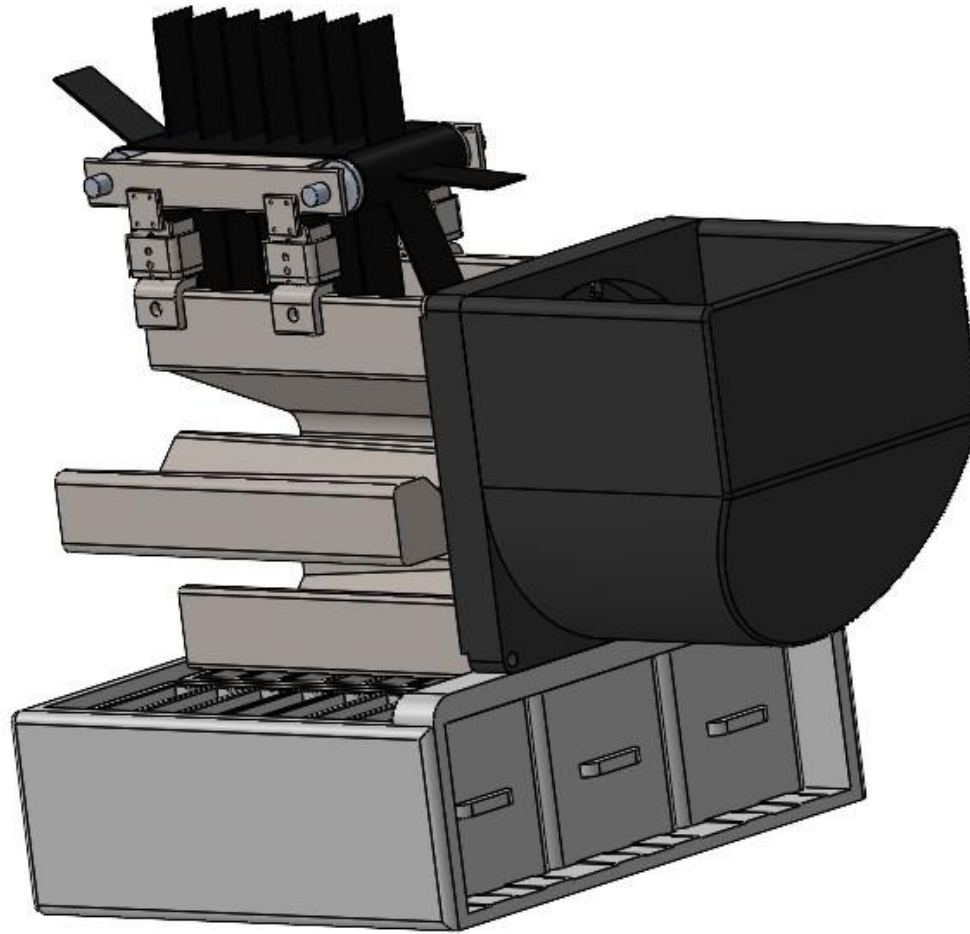
7. Make the hole of the inner wheel concentric with the rod and press the inner wheel into it with a tight fit until the inner wheel's face is touching the outer wheel's face. Leave a small portion of the rod sticking out on both ends.
8. Create space for the collar to be attached to the end of the chamfered side of the rod. Align the outer edge of the collar with the beginning of the chamfer and press it against the face of the inner wheel. Tighten using a small screw and insert it into the two holes on the collar.
9. Align the funnel so that the open side is facing the inner wheel and the bottom curved side is facing downward. Look to make the small holes at the top of the outer wheel and the top edges of the funnel concentric to each other and connect with a screw.



10. For each slide, use the end with the highest elevation and align the edge of the slide with the edge of the hole on the outer wheel. Ensure the midpoints of each edge are aligned with one another, and then weld them to the outer wheel.
11. Attach the motor to the hole in the small rod used to hold the inner wheel in place. Use adhesive to hold these two together.
12. Align the conveyor subassembly with the outer edges of the bolt slide. Ensure that the faces of the belt foot and the outside face of the bolt slide are parallel, and use a screw to attach them to the sides of the bolt slide.



13. The complete subassembly should resemble this.



Assembly Instruction Manual: Chassis Subassembly (Naitik Mundra)

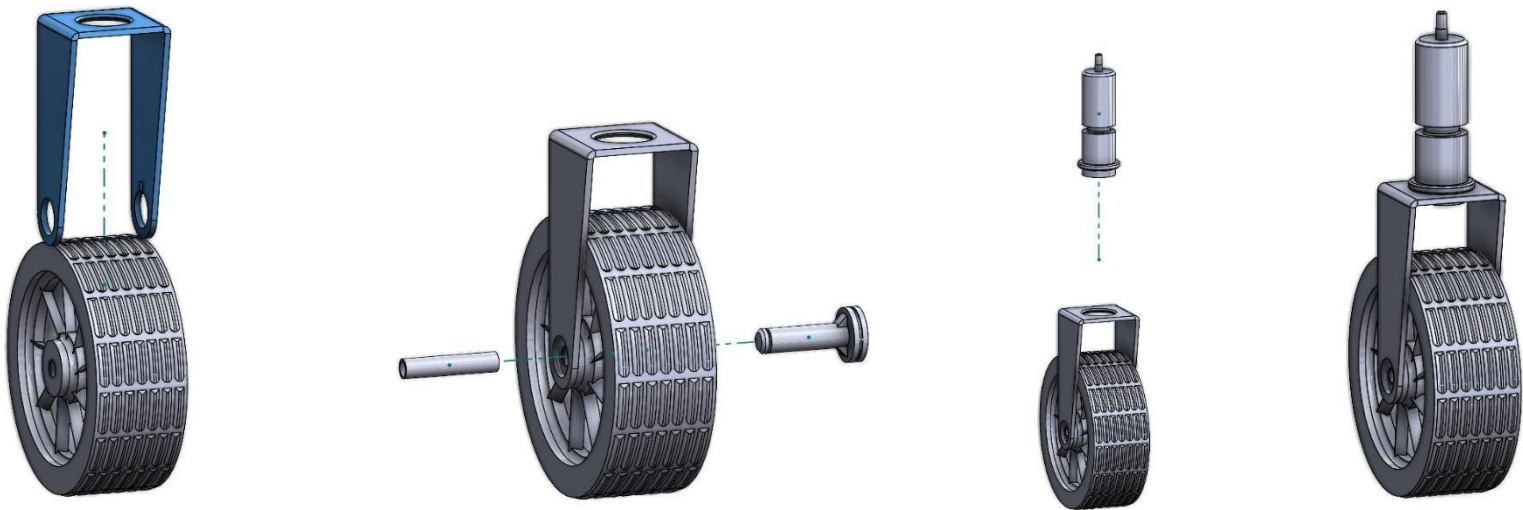
There are 3 separate sub-subassemblies that bolt on to the main frame to make the main subassembly. These are:

- Wheels
- Vertical Elevator
- Branch

□ This has one further subassembly: the Pusher

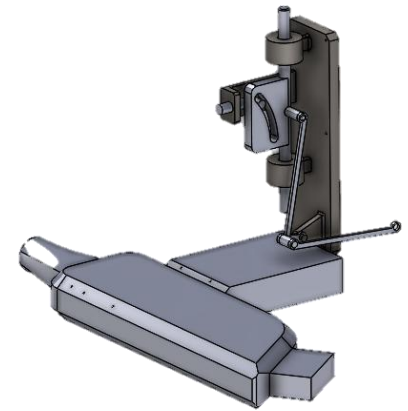
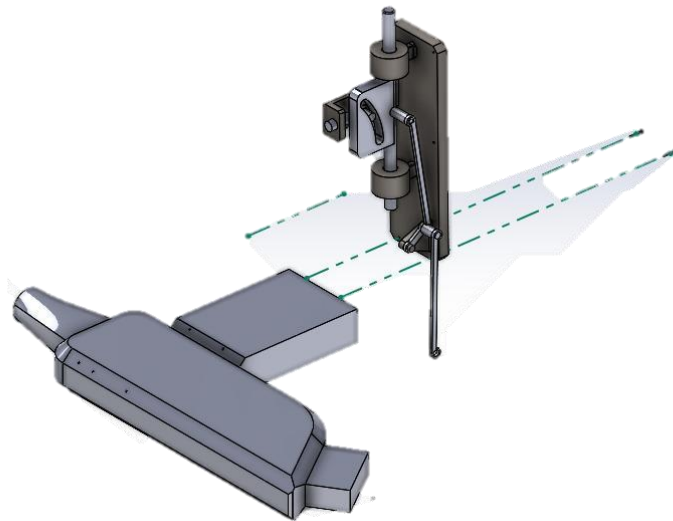
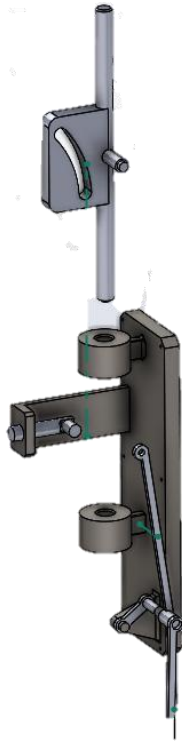
Wheels

1. Place the bracket on top of the wheel, using the bolt and bushing. Make sure to insert the bolt oriented such that the locking clip (visible on top of the bolt in the diagram below) locks into place in the cavity provided on the wheel rim.
2. Fix the posting onto the bracket with a simple press into a clearance fit, making sure to not overtighten too much – the wheel needs to swivel!



Branch

1. Make the pusher by attaching the cam and crank mechanisms as shown in the diagram
2. Use binding head screws to attach the branch to the base of the pusher



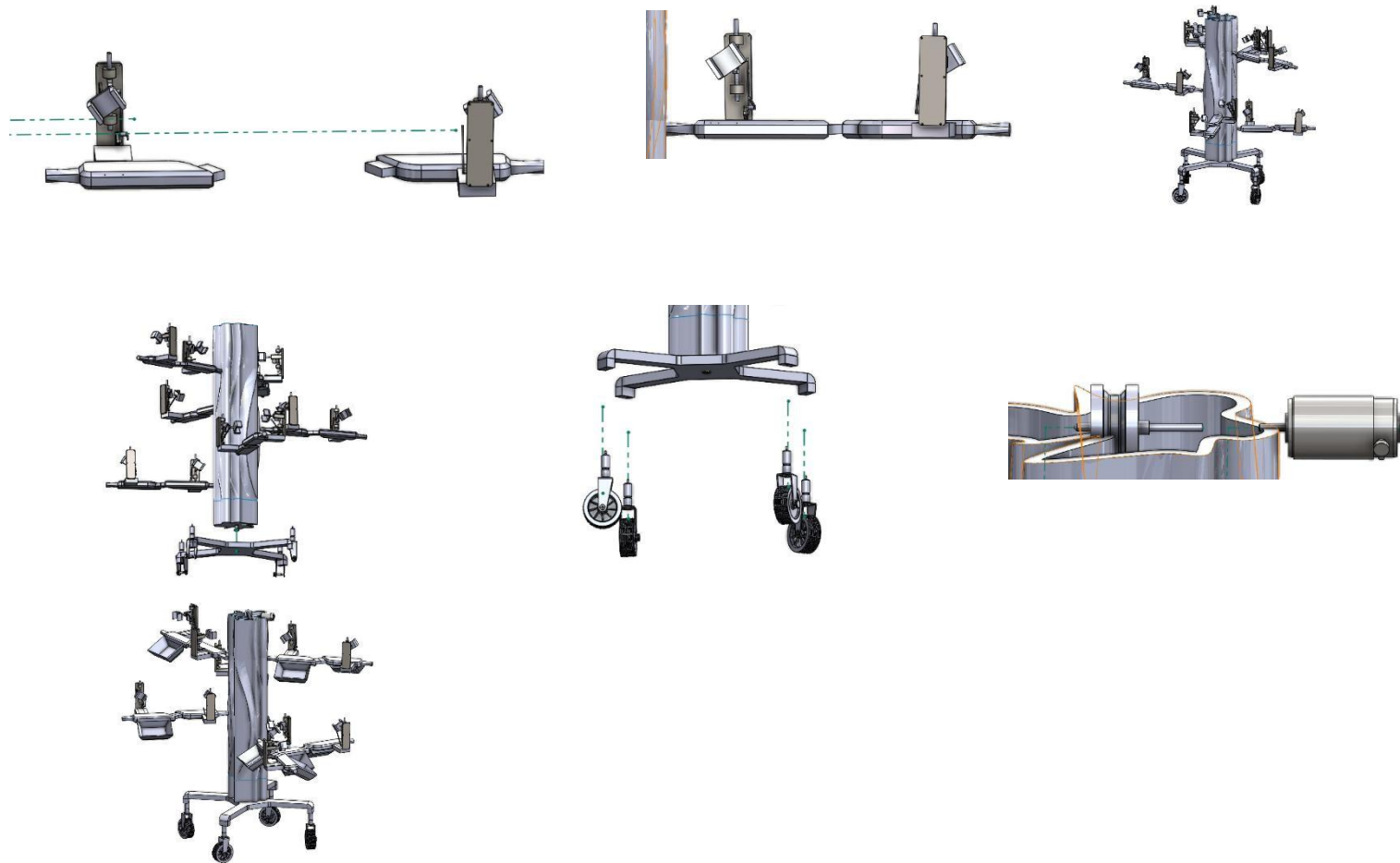
Vertical Elevator

1. Wrap a rope around the pulley and attach it to the hook.
2. Run the axle through the sheave



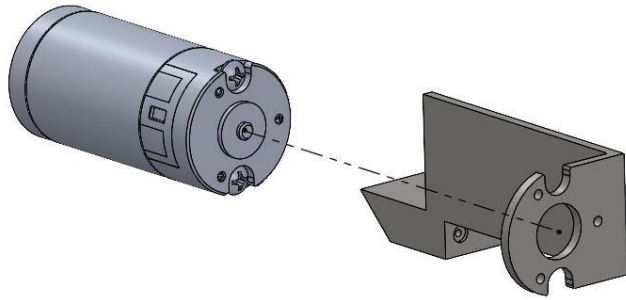
Main Chassis Assembly

1. Attach two branches together to create a single level Do this with welding, and place it so pushers face opposite sides to increase amount of space available
2. Hollow out a small piece of the main trunk with a hole saw and then press fit the circular portion of the first branch into the tree. Make sure to have some overlap inside so the hook can place objects into conveyors rather than moving aimlessly.
3. Do the above branch connection for all branches in a radial pattern
4. Attach a cross bracket to the bottom of the trunk
5. Connect all 4 wheels to the cross bracket
6. Attach a motor to the axle of vertical elevator and then wedge it onto the shelled portion of the stem
7. Weld the buckets on. This step should be done last to ensure clearance. If done along with branch assemblies, the entire manufacturing process becomes cumbersome. Hence, done last.



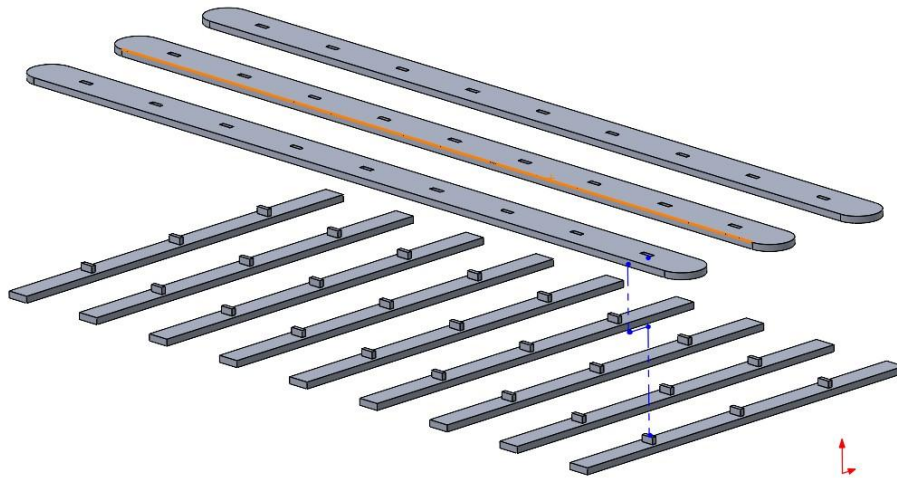
Assembly Instruction Manual: Motor Mount Subassembly (Calvin Arnold)

1. Align motor with motor mount

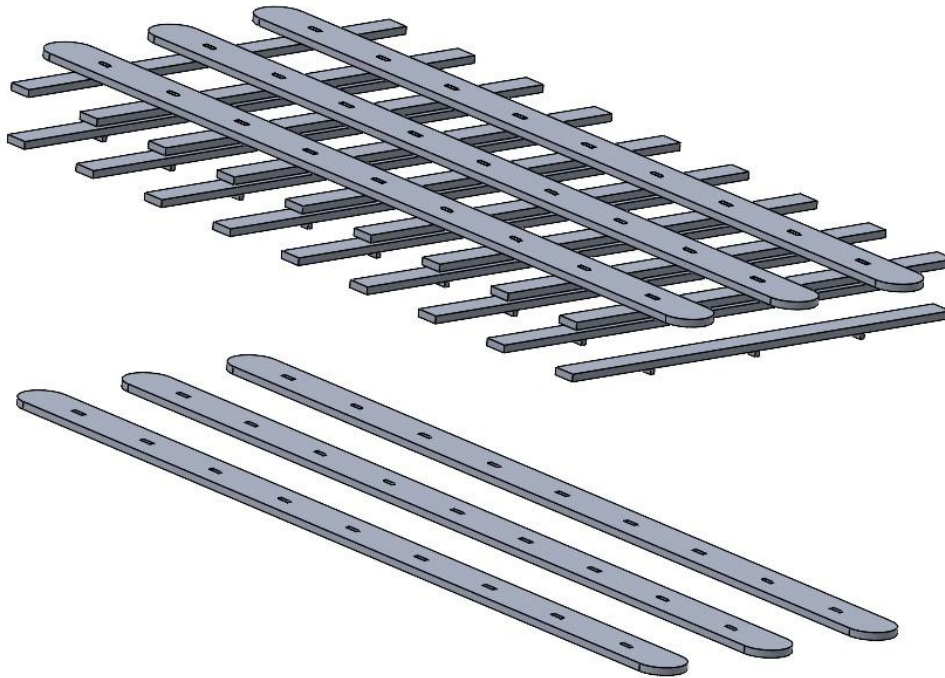


Assembly Instruction Manual: Conveyor Subassembly (Calvin Arnold)

1. Join the horizontal guide support with the conveyor support, aligning the holes



2. Repeat this step again, orienting the conveyor supports facing down.



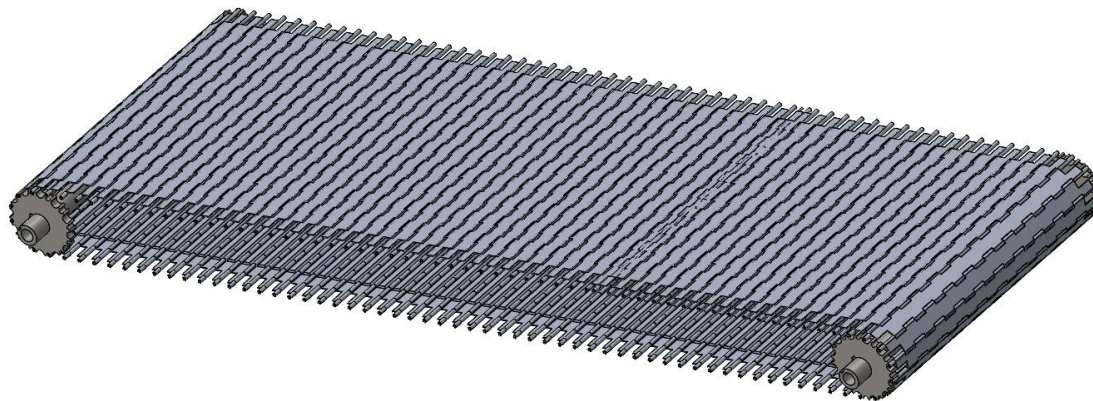
3. Slot axel pin into axel, centering both.
4. Repeat step 3



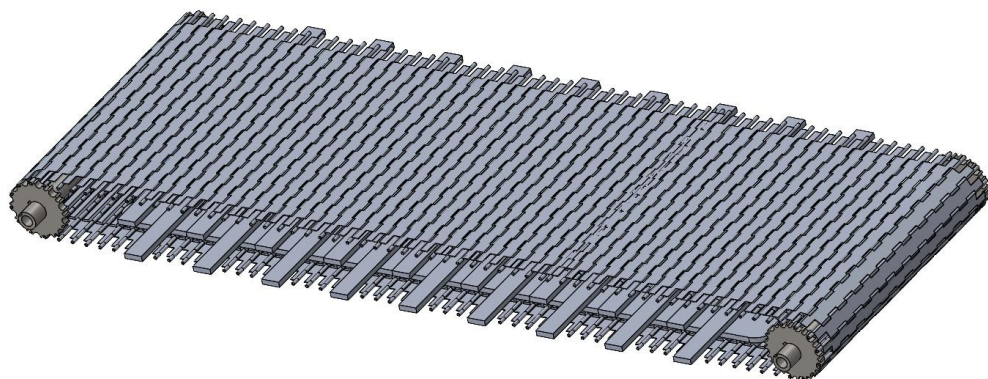
5. Align sprocket with each axel, one on each end. Repeat 4x.



6. Join each link together, all 120 instances, around the sprockets.



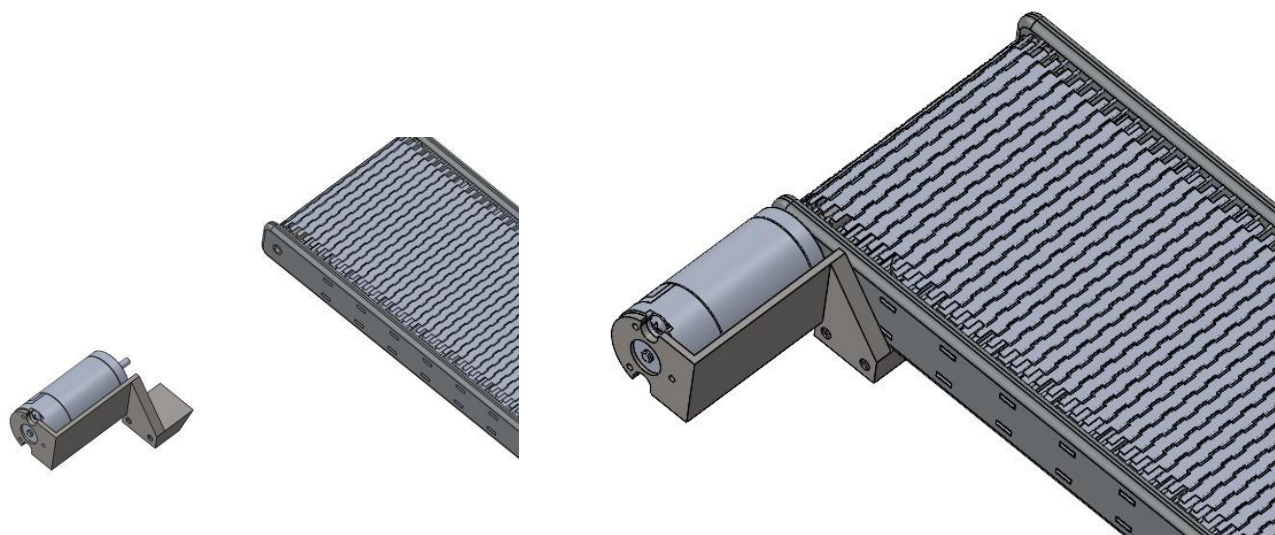
7. Slot the supports from steps 1-2 in the center of the conveyor to support it.



8. Align the conveyor guides with the axels and guide supports, slotting each into the designated holes.



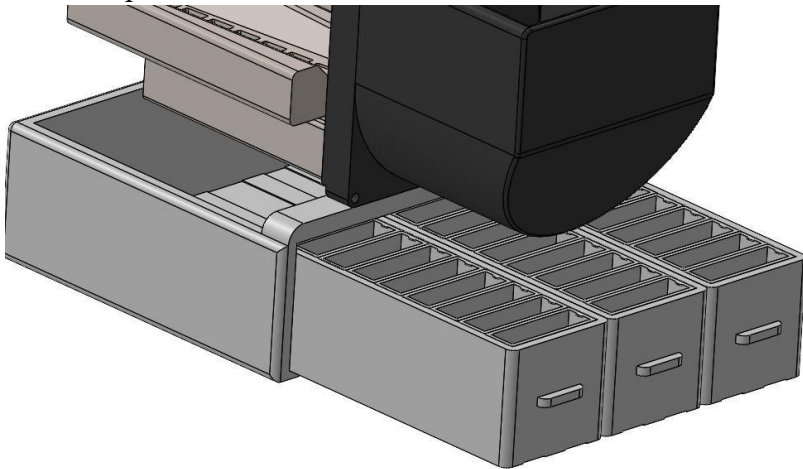
9. Align Motor subassembly with the bottom of the conveyor guides, putting the shaft of the motor within the slot on the axel



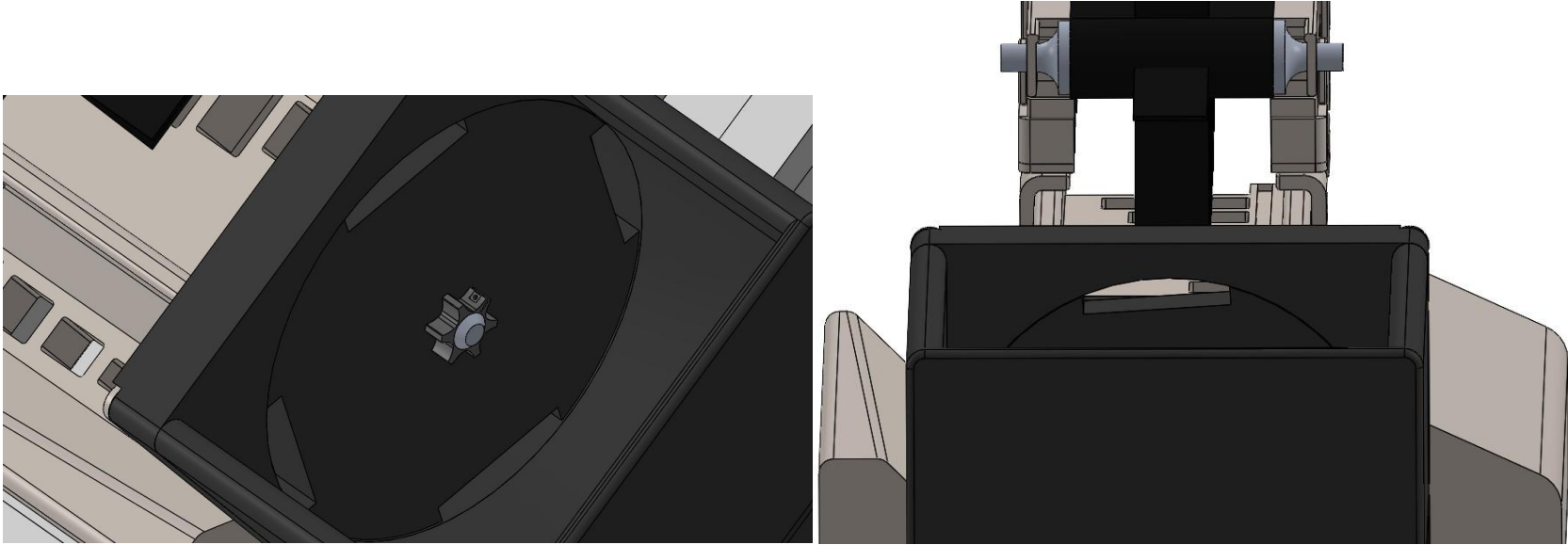
Chapter 5

Check for Functionality

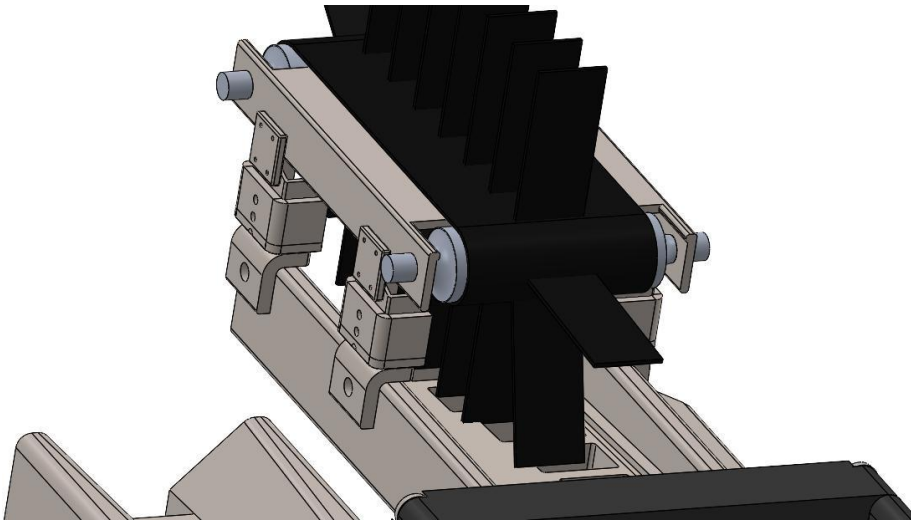
The boxes in the outer container can slide out to easily access the sorted fasteners. Within these, the small boxes can also be slid out to access a specific size.



The inner wheel rotates to move the fasteners deposited into the funnel into the correct opening as shown in the second image. This wheel constantly rotates at a slow rate to ensure that the shape of each fastener type fits into its corresponding hole and is sorted in the slide by gravity. The collar prevents the wheel from slipping out.



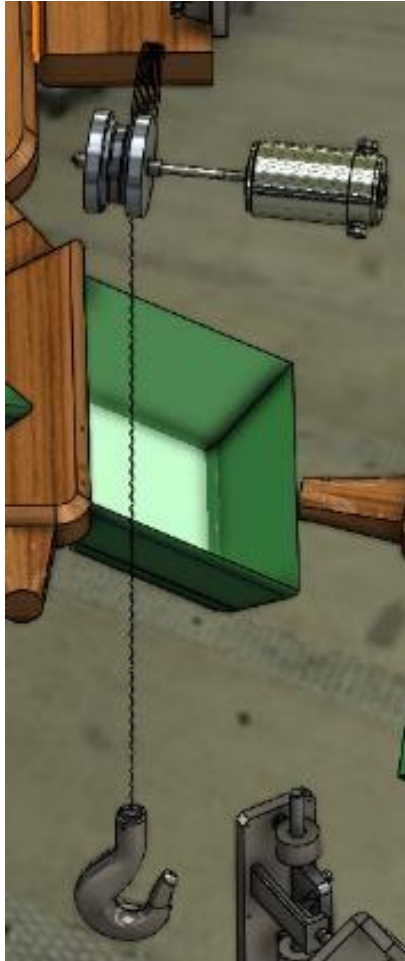
The conveyor turns with the rotation of the cylinders, which are powered by a motor inside the belt holder. Each extrusion on the conveyor moves alongside it and brushes against the surface of the slide to push bolts along the track.



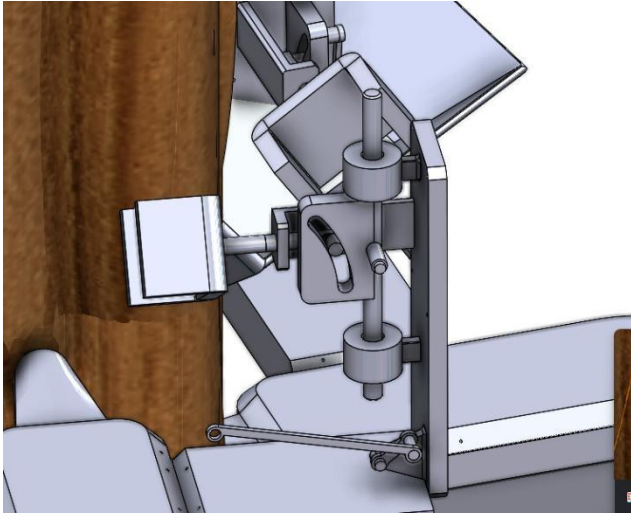
The stem/trunk of Toolio is situated on 4 castor wheels placed symmetrically through the use of a cross bracket, which itself has a clearance fit to allow for the main body to swivel 360 degrees for ease of access to tools.



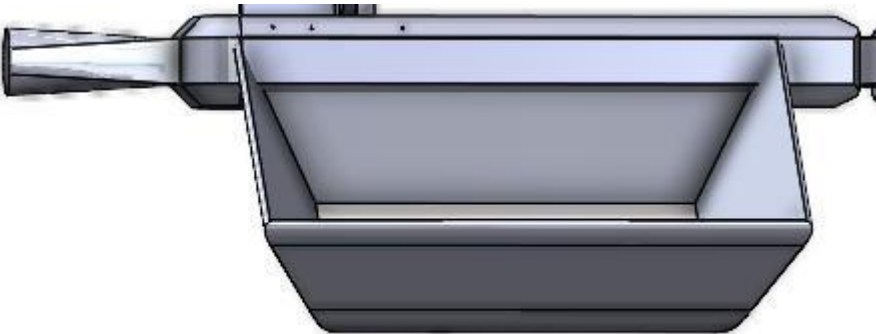
When an object is placed onto the hook, the pulley and rope system comes into action, using the DC-motor to lower the object to the correct height



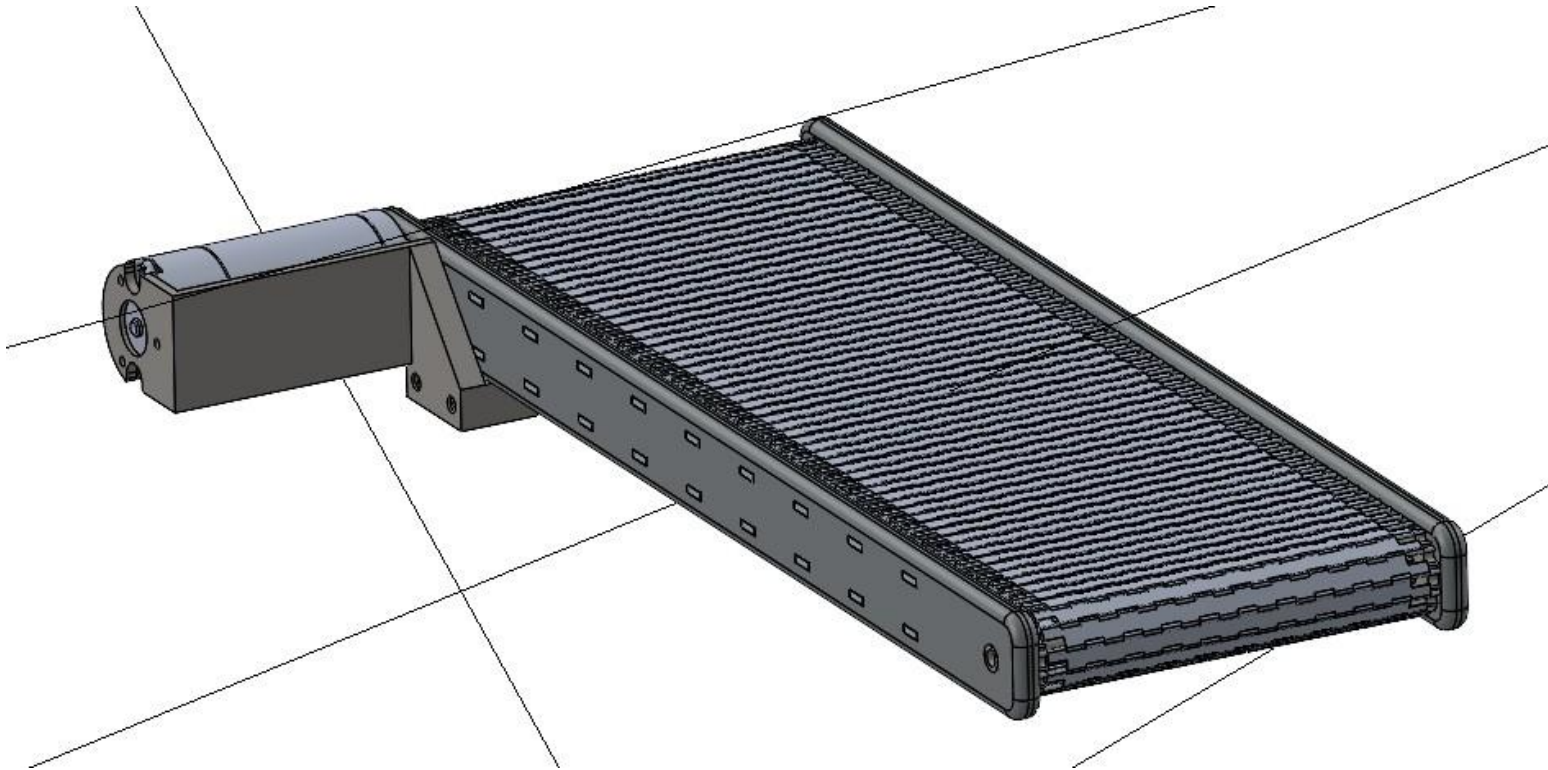
Then, from the conveyor, the pusher system can push into baskets.



The baskets are also movable themselves to allow for some flexibility



The conveyor is able to spin around itself, allowing the objects placed on top to be moved horizontally down the branches.



Chapter 6

Summary and Concluding Remarks

This project started out with a seemingly simple objective: to make an easy-to-use sorting system for tools. But we did not realise the number of complications that could arise, and the number of moving parts needed to make it a better user experience. Our initial designs were focused on a single conveyor belt that would just distribute items as needed. However, we realized that this is not all that is needed. The problem specifically asks us to be mindful of the fact that this solution needs to be easier to use and more productive than regular toolboxes. A simple conveyor belt relying solely on geometry to get the job done was asking for trouble – it could get stuck, and this would waste valuable time. We also realized that the most annoying part of this could be sorting fasteners such as bolts, nuts, screws etc. Hence, we also decided to add this as a feature. This gradual iteration taught us the value of an iterative process. One can even see this iteration from the evolution of our sketches to CAD models. The branch subassembly went from just a holder for parts to something with over half a dozen complications and even a cam mechanism to accommodate these changes. However, the initial designs and sketches were still valuable, since they allowed us to focus on the improvements to be made in an incremental fashion. This further reinforced our learning about how a real capstone project might be designed, and the various ways in which a more experienced engineer might go about designing a systemic solution to real customer requirements.

Our time spent with the mentor understanding various parts of the design was also helpful. The initial plan to have a complex spring based locking system like a seatbelt for the wheels seemed ambitious – she was able to suggest the much better alternative of just a clip with a tight clearance fit. She also reassured us of our designs and gave us guidance on how to know whether we were on the right track.

Simultaneously, this was our first time managing a CAD project across multiple teammates in such a formal fashion. This causes significant friction in how we approached collaborative editing, especially with things such as assemblies and animations. Nevertheless, we now feel much more confident about our skills going forward. Based on all of these factors, we can conclude that this project was a significant success in our learning journey of exploring the design process from ideation till the final parts of the design stage, and we are excited to continue to learn about the next part of the product life cycle such as DFM and more.